SCL3300

Data Sheet

SCL3300-D01
3-axis inclinometer with angle output and digital SPI interface



Features

- 3-axis (XYZ) inclinometer
- User selectable measurement modes:
  - 3000 LSB/g with 70 Hz LPF
  - 6000 LSB/g with 40 Hz LPF
  - 12000 LSB/g with 10 Hz LPF
- Angle output resolution 0.0055°/LSB
- −40°C…+125°C operating range
- 3.0V…3.6V supply voltage
- SPI digital interface
- Ultra-low 0.001 °/√Hz noise density
- Excellent offset stability
- Size 8.6 x 7.6 x 3.3 mm (l × w × h)
- Proven capacitive 3D-MEMS technology

Applications

SCL3300-D01 is targeted at applications demanding high stability and accuracy with tough environmental requirements.

Typical applications include:

- Leveling
- Tilt sensing
- Machine control
- Structural health monitoring
- Inertial measurement units (IMUs)
- Robotics
- Positioning and guidance systems

Overview

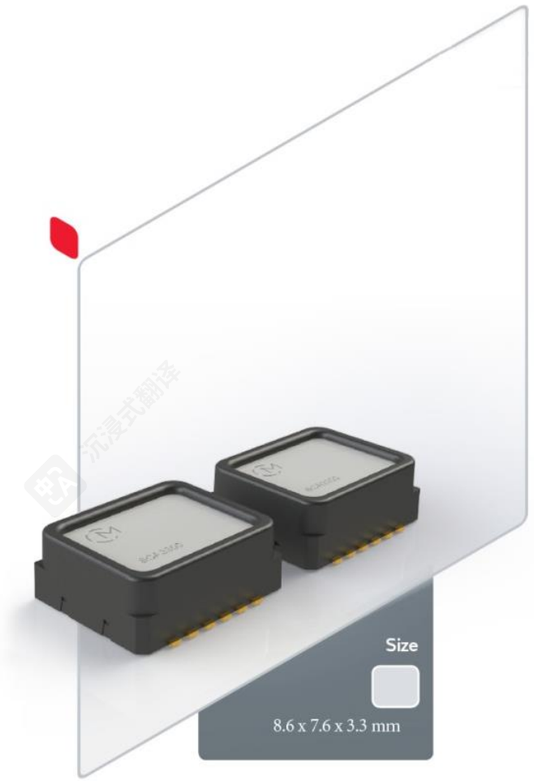
The SCL3300-D01 is a high performance inclinometer sensor component. It is a three-axis inclinometer sensor with angle output based on Murata's proven capacitive 3D-MEMS technology. Signal processing is done in a mixed signal ASIC with flexible SPI digital interface. Sensor element and ASIC are packaged into 12 pin pre-molded plastic housing that guarantees reliable operation over product's lifetime.

The SCL3300-D01 is designed, manufactured and tested for high stability, reliability and quality requirements. The component has extremely stable output over wide range of temperature and vibration. The component has several advanced self-diagnostics features, is suitable for SMD mounting and is compatible with RoHS and ELV directives.

SCL3300

数据表

SCL3300-D01
带角度输出的三轴倾角仪 数字 SPI 接口



功能特性

- 三轴（XYZ）倾角仪
- 用户可选择的测量模式：
  - 3000 LSB/g，70 Hz 低通滤波器
  - 6000 LSB/g，40 Hz 低通滤波器
  - 12000 LSB/g，10 Hz 低通滤波器
- 角度输出分辨率 0.0055°/LSB
- −40°C…+125°C 工作范围
- 3.0V...3.6V 供电电压
- SPI 数字接口
- 超低 0.001 °/√Hz 噪声密度
- 卓越的偏移稳定性
- 尺寸 8.6 x 7.6 x 3.3 毫米（长 × 宽 × 高）
- 经过验证的电容式 3D-MEMS 技术

应用

SCL3300-D01 专为需要高性能的应用而设计在严苛的环境要求下保持稳定性和准确性。

典型应用场景包括：

- 等级
- 倾斜感应
- 机器控制
- 结构健康监测
- 惯性测量单元 (IMU)
- 机器人
- 定位与导航系统

概览

SCL3300-D01 是一款高性能倾角传感器组件。它是一款三轴倾角传感器，基于 Murata 成熟的电容式 3D-MEMS 技术输出角度信号。信号处理采用混合信号 ASIC 完成，并配备灵活的 SPI 数字接口。传感器元件与 ASIC 封装于 12 引脚预成型塑料外壳中，确保在整个产品生命周期内可靠运行。

SCL3300-D01 专为高稳定性、高可靠性和高质量要求而设计、制造和测试。该元件在宽温范围和振动条件下具有极其稳定的输出。该元件具备多项先进的自诊断功能，适用于 SMD 封装，并符合 RoHS 和 ELV 指令要求。

TABLE OF CONTENTS

1 Introduction 4

2 Specifications 4

2.1 Abbreviations 4

2.2 General Specifications 4

2.3 Performance Specifications for Inclinometer 5

2.4 Performance Specification for Temperature Sensor 7

2.5 Specification for Angle Outputs 7

2.6 Absolute Maximum Ratings 8

2.7 AEC-Q100 Testing 8

2.8 Pin Description 9

2.9 Typical performance characteristics 10

2.10 Digital I/O Specification 13

2.10.1 SPI DC Characteristics 13

2.10.2 SPI AC Characteristics 14

2.11 Measurement Axis and Directions 15

2.11.1 Measurement Ranges on Inclination Modes 16

2.12 Package Characteristics 17

2.12.1 Package Outline Drawing 17

2.13 PCB Footprint 18

3 General Product Description 19

3.1 Factory Calibration 19

4 Component Operation, Reset and Power Up 20

4.1 Component Operation 20

4.2 Start-up sequence 21

4.3 Operation modes 22

5 Component Interfacing 22

5.1.1 General 22

5.1.2 Protocol 23

5.1.3 SPI frame 24

5.1.4 Operations 25

5.1.5 Return Status 26

5.2 Checksum (CRC) 26

6 Register Definition 27

6.1 Sensor Data Block 29

6.1.1 Example of Acceleration Data Conversion 30

6.1.2 Example of Temperature Data Conversion 31

目录

1 介绍 ction .4

2 Sp规范 **4**

2.1 缩写 viatio ns .4

2.2 Gen eral S 规格 .4

2.3 倾角仪性能规范.5

2.4 温度传感器性能规范.7

2.5 SpA 规范 ngle 输出 uts .7

2.6 Absolute 最大 mum Ratin gs .8

2.7 AEC-Q1 00 T esting .8

2.8 引脚 D escri选项 .9

2.9 T典型表现 种质特征.10

2.10 Dig ital I/O S 规格 .1 3

2.10.1 SPI 直流特性. 13

2.10.2 SPI 交流特性 14

2.11 测量 m主轴 和方向.15

2.11.1 倾斜模式下的测量范围.16

2.12 包 特性 17

2.12.1 Package 大纲 D绘图 17

2.13 PCB Footprint 18

3 通用产品 产品描述.19

3.1 Fa胜利校准 n .1 9

4 组件操作、复位与上电.20

4.1 Comp onent O 操作 .20

4.2 启动 se序列 21

4.3 Op运行 模式 22

5 Comp组件接口 g.22

5.1.1 生成 l 22

5.1.2 Protoco l 23

5.1.3 SPI 从mė 24

5.1.4 Ope rations 25

5.1.5 Re切换状态 s .2 6

5.2 检查 m (CRC) .2 6

6 Reg继承定义 .27

6.1 Se nsor D ata 块 ck 29

6.1.1 Exam加速数据转换示例.30

6.1.2 Exam温度数据转换示例.31

6.1.3 Example of Angle Data Conversion 32
6.2 STO 33
6.2.1 Example of Self-Test Analysis 34
6.3 STATUS 35
6.3.1 Example of STATUS summary reset 37
6.4 Error Flag Block 38
6.4.1 ERR_FLAG1 38
6.4.2 ERR_FLAG2 39
6.5 CMD 40
6.6 ANG_CTRL 41
6.7 WHOAMI 41
6.8 Serial Block 42
6.8.1 Example of Resolving Serial Number 43
6.9 SELBANK 44
7 Application information 44
7.1 Application Circuitry and External Component Characteristics 44
7.2 Assembly Instructions 46
8 Frequently Asked Questions 46
9 Order Information 47

6.1.3 角度数据转换示例.32
6.2 STO 33
6.2.1 Exam自我 - 利 是A 分析 34
6.3 STATUS 35
6.3.1 Exam状态摘要重置.37
6.4 错误 Flag Block 38
6.4.1 错误 _FLAG1 .3 8
6.4.2 ERR FLA_ G2 .3 9
6.5 CMD . .40
6.6 ANG CT_ RL .41
6.7 WHOAMI 41
6.8 Serial Block. 42
6.8.1 Exam解决串行 N 的示例 um伯.43
6.9 SELBAN K. 44
7 A应用信息 ation .44
7.1 应用电路与外部元件特性.44
7.2 Assembly 说明 ns 46
8 经常 常见问题.46
9 Order 信息 ation . .47

1 Introduction

This document contains essential technical information about the SCL3300-D01 sensor including specifications, SPI interface descriptions, user accessible register details, electrical properties and application information. This document should be used as a reference when designing in SCL3300-D01 component.

2 Specifications

2.1 Abbreviations

ASIC	Application Specific Integrated Circuit
SPI	Serial Peripheral Interface
RT	Room Temperature, +23 °C
FS	Full Scale
CSB	Chip Select
SCK	Serial Clock
MOSI	Master Out Slave In
MISO	Master In Slave Out
MCU	Microcontroller
STO	Self-test Output

2.2 General Specifications

General specifications for SCL3300-D01 component are presented in Table 1. All analog voltages are related to the potential at AVSS and all digital voltages are related to the potential at DVSS.

Table 1 General specifications

Parameter	Condition	Min	Nom	Max	Units
Supply voltage: VDD		3.0	3.3	3.6	V
SPI supply voltage: DVIO	Must never be higher than VDD	3.0	3.3	3.6	V
Current consumption: I_VDD	Temperature range -40 ... +125 °C Standard operation Mode 4		1.2 2.1		mA
Current consumption: I_VDD in power down mode	Temperature range -40 ... +125 °C Power down mode (PD) Typical value is at room temperature (+23°C)		3	10	µA

1 简介

本文档包含关于 SCL3300-D01 传感器的关键技术信息，包括规格、SPI 接口描述、用户可访问的寄存器详情、电气特性及应用信息。在设计 SCL3300-D01 组件时，应参考本文档。

2 规格

2.1 缩写

ASIC	专用集成电路
SPI	串行外设接口
RT	室温，+23 °C
FS	全尺寸
CSB	芯片选择
SCK	串行时钟
MOSI	主输出 从输入
MISO	主从输出
MCU	微控制器
STO	自我测试输出

2.2 一般规格

SCL3300-D01 组件的通用规格见表 1。所有模拟电压均相对于 AVSS 电位，所有数字电压均相对于 DVSS 电位。

表 1 通用规格

参数	条件	Min	Nom	Max	单位
供电电压：VDD		3.0	3.3	3.6	V
SPI 供电电压：DVIO	必须始终低于 VDD	3.0	3.3	3.6	V
当前功耗：I_VDD	温度范围 -40 … +125 °C 标准操作 模式 4		1. 		mA
当前功耗：I_VDD 在掉电模式下	温度范围 -40 … +125 °C 关机模式 (PD) 典型值是在室温下 (+23°C)		3	10	µA

2.3 Performance Specifications for Inclinometer

Table 2 Inclinometer performance specifications. Supply voltage VDD = 3.3 V and room temperature (RT) +23 °C unless otherwise specified. Definition of gravitational acceleration: g = 9.819 m/s².

Parameter	Condition	Min	Nom	Max	Unit
Measurement range	Mode 1 Mode 2 Mode 3, Mode 4(A		1.2 2.4 -		g
	Mode 1 Mode 2 Mode 3, Mode 4(A		±90 ±90 (±10)		°
Offset error (B	All modes, X, Z channels	-20 -1.15		20 1.15	mg °
	All modes, Y channel	-25 -1.45		20 1.15	mg °
Offset temperature dependency (C	-40°C ... +125°C, X, Y channels	-10 -0.57		10 0.57	mg °
	-40°C ... +125°C, Z channel	-15 -0.86		15 0.86	mg °
Offset lifetime drift (D	All modes, X, Z channels	-8 -0.46	±4 ±0.23	8 0.46	mg °
	All modes, Y channels	-12 -0.69	±6 ±0.34	12 0.69	mg °
Sensitivity (acceleration output)	Mode 1 Mode 2 Mode 3, Mode 4		6000 3000 12000		LSB/g
	Mode 1 Mode 2 Mode 3, Mode 4 valid only between 0…1° (E		105 52 209		LSB/°
Sensitivity (inclination output)	All modes		182		LSB/°
Sensitivity error (B	-40°C ... +125°C Mode 1	-0.7		0.7	%
Sensitivity temperature dependency (C	-40°C ... +125°C Mode 1	-0.3		0.3	%
Linearity error (F	-1g ... +1g range	-4		4	mg
Integrated noise (RMS, accelerometer) (G	Mode 3, X, Y, Z channels Mode 4, X, Z channels Mode 4, Y channel		0.13 0.08 0.06		mg_{RMS}
Noise density (G	Mode 3, X, Y, Z channels Mode 4, X, Z channels Mode 4, Y channel		32 20 15		µg/√Hz
	Mode 3, X, Y, Z channels Mode 4, X, Z channels Mode 4, Y channel		0.0018 0.0012 0.0009		°/√Hz
Cross axis sensitivity (H	per axis	-1.5	±0.2	1.5	%
Amplitude response, -3dB frequency	Mode 1		40		Hz
	Mode 2		70		Hz
	Mode 3, Mode 4		10		Hz

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2.3 倾角仪性能规格

表 2 倾斜仪性能规格。供电电压 VDD = 3.3 V，除非另有说明，否则在室温 (RT) +23 °C 下。重力加速度定义：g = 9.819 m/s²。

参数	条件	Min	Nom	Max	Unit
测量范围	模式 1 模式 2 模式 3 ，模式 4(A		1.2 2.4 -		g
	模式 1 模式 2 模式 3，模式 4(A		±90 ±90 (±10)		°
偏移误差（B	所有模式，X、Z 通道	-20 -1.15		20 1.15	mg °
	所有模式，Y 通道	-25 -1.45		20 1.15	mg °
偏移温度依赖性（C	-40°C ... +125°C，X，Y 通道	-10 -0.57		10 0.57	mg °
	-40°C ... +125°C，Z 通道	-15 -0.86		15 0.86	mg °
偏移寿命漂移（D	所有模式，X、Z 通道	-8 -0.46	±4 ±0.23	8 0.46	mg °
	所有模式，Y 通道	-12 -0.69	±6 ±0.34	12 0.69	mg °
灵敏度（加速度输出）	模式 1 模式 2 模式 3 ，模式 4		6000 3000 12000		LSB/g
	模式 1 模式 2 模式 3 ，模式 4 仅在 0...1°（E 之间有效		105 52 209		LSB/°
灵敏度（倾角输出）	所有模式		182		LSB/°
灵敏度错误 (B	-40°C ... +125°C 模式 1	-0.7		0.7	%
灵敏度温度依赖 (C	-40°C ... +125°C 模式 1	-0.3		0.3	%
线性误差 (F	-1g … +1g 范围	-4		4	mg
集成噪声（RMS，加速度计) (G	模式 3，X、Y、Z 通道 模式 4，X、Z 通道 模式 4 ，Y 通道		0.13 0.08 0.06		mg_{RMS}
噪声密度 (G	模式 3 ，X、Y、Z 通道 模式 4，X、Z 通道 模式 4 ，Y 通道		32 20 15		µg/√Hz
	模式 3 ，X、Y、Z 通道 模式 4，X、Z 通道 模式 4 ，Y 通道		0.0018 0.0012 0.0009		°/√Hz
横轴灵敏度（H	每轴	-1.5	±0.2	1.5	%
幅频响应，-3dB 频率	模式 1		40		Hz
	模式 2		70		Hz
	模式 3，模式 4		10		Hz

Parameter	Condition	Min	Nom	Max	Unit
Power on start-up time[I]	Mode 1		25		ms
	Mode 2		15		ms
	Mode 3, Mode 4		100		ms
ODR			2000		Hz

Min/Max values are ±3 sigma variation limits from test population at the minimum. Min/Max values are not guaranteed.

A) Inclination mode. Dynamic range is dependent on orientation in gravity. See Chapter 2.11.1

B) Includes calibration error, temperature, supply voltage and drift over lifetime.

C) Deviation from value at room temperature (RT).

D) Min/Max results based on the maximum ±3 sigma variation limits of the following tests:

HTOL bake (+125 °C, 1000h), TC (-50 °C / +150 °C, 1000 cycles), THB (85 °C / 85 %RH).

E) Angle calculated using 1g * SIN(θ), where θ is the inclination angle relative to the 0g position. Due to characteristics of sine function sensitivity is inversely proportional to inclination angle. Reported values are valid only between 0° to ±1°.

F) Straight line through specified measurement range end points.

G) SPI communication may affect the noise level. Used SPI clock should be carefully validated. Recommended SPI clock is 2 MHz - 4 MHz to achieve the best performance; see section 2.10.2 SPI AC Characteristics for details.

H) Cross axis sensitivity is the maximum sensitivity in the plane perpendicular to the measuring direction. X-axis output cross axis sensitivity (cross axis for Y and Z-axis outputs are defined correspondingly):

- Cross axis for Y axis = Sensitivity Y / Sensitivity X
- Cross axis for Z axis = Sensitivity Z / Sensitivity X

I) Power on start-up time is specified according to recommended start-up sequence; see section 4.2 Start-up sequence for details.

参数	条件	Min	Nom	Max	Unit
上电启动时间[I]	模式 1		25		ms
	模式 2		15		ms
	模式 3，模式 4		100		ms
ODR			2000		Hz

最小/最大值为测试群体中最低点的±3 σ变异极限。最小/最大值不保证。

A) 倾斜模式。动态范围取决于重力方向。参见第 2.11.1 节。

B) 包含校准误差、温度、供电电压和全寿命周期内的漂移。

C) 与室温（RT）下数值的偏差。

D) 基于以下测试的最大 ±3 sigma 变化极限的 Min/Max 结果：

HTOL 烘烤（+125 °C，1000 小时），TC（-50 °C / +150 °C，1000 次循环），THB（85 °C / 85 %RH）。

E) 使用 1g * SIN(θ) 计算角度，其中 θ 是相对于 0g 位置的倾斜角。由于正弦函数的特性，其灵敏度与倾斜角成反比。报告的值仅在 0° 到 ±1° 之间有效。

F) 通过指定测量范围端点的直线。

G) SPI 通信可能会影响噪声电平。应仔细验证所使用的 SPI 时钟频率。为实现最佳性能，推荐的 SPI 时钟频率为 2 MHz 至 4 MHz；详见 2.10.2 节"SPI AC 特性”。

H) 正交轴灵敏度是垂直于测量方向的平面内的最大灵敏度。
X 轴输出交叉轴灵敏度（Y 轴和 Z 轴输出的交叉轴定义相应）：

- Y 轴交叉轴 = 灵敏度 Y / 灵敏度 X
- Z 轴交叉轴 = 灵敏度 Z / 灵敏度 X

I) 上电启动时间根据推荐的启动顺序指定；详见第 4.2 节“启动顺序”。

2.4 Performance Specification for Temperature Sensor

Table 3 Temperature sensor performance specifications.

Parameter	Condition	Min.	Typ	Max.	Unit
Temperature signal range		-50		+150	°C
Temperature signal sensitivity	Direct 16-bit word		18.9		LSB/°C
Temperature signal offset	°C output	-10		10	°C

Temperature is converted to °C with following equation:

```
Temperature [°C] = -273 + (TEMP / 18.9),
```

where TEMP is temperature sensor output register content in decimal format.

2.5 Specification for Angle Outputs

Angles are formed from acceleration with following equations:

```
ANG_X = atan2(accx / √(accy^2 + accz^2)),
ANG_Y = atan2(accy / √(accx^2 + accz^2)),
ANG_Z = atan2(accz / √(accx^2 + accy^2)),
```

where accx, accy, and accz are accelerations to each direction and ANG_X, ANG_Y, and ANG_Z are angle output register content in 16-bit binary format.

Angles are converted to degrees with following equation:

```
Angle [°] = d'ANG_% / 2^14 * 90,
```

where d'ANG_% is angle output register (ANG_X, ANG_Y, ANG_Z) content in decimal format. See 6.1.3 Example of Angle Data Conversion for more information.

2.4 温度传感器的性能规范

表 3 温度传感器性能规格。

参数	条件	Min.	Typ	Max.	Unit
温度信号范围		-50		+150	°C
温度信号灵敏度	直接 16 位字		18.9		LSB/°C
温度信号偏移	°C 输出	-10		10	°C

温度通过以下公式转换为摄氏度：

```
Temperature [°C] = -273 + (TEMP / 18.9),
```

其中 TEMP 为温度传感器输出寄存器的十进制内容。

2.5 角度输出规范

角度由加速度通过以下方程形成：

```
ANG_X = atan2(accx / √(accy^2 + accz^2)),
ANG_Y = atan2(accy / √(accx^2 + accz^2)),
ANG_Z = atan2(accz / √(accx^2 + accy^2)),
```

其中 accx、accy 和 accz 分别表示各方向的加速度，ANG_X、ANG_Y 和 ANG_Z 为角度输出寄存器内容，采用 16 位二进制格式。

角度按以下公式转换为度：

```
Angle [°] = d'ANG_% / 2^14 * 90,
```

其中 d'ANG_% 为角度输出寄存器（ANG_X、ANG_Y、ANG_Z）中的十进制内容。有关更多信息，请参阅 6.1.3 角度数据转换示例。

2.6 Absolute Maximum Ratings

Within the maximum ratings (Table 4), no damage to the component shall occur. Parametric values may deviate from specification, yet no functional failure shall occur.

Table 4. Absolute maximum ratings.

Symbol	Description	Min.	Typ	Max.	Unit
VDD	Supply voltage analog circuitry	-0.3		4.3	V
DIN/DOUT	Maximum voltage at digital input and output pins	-0.3		DVIO+0.3	V
Topr	Operating temperature range	-40		+125	°C
Tstg	Storage temperature range	-40		+150	°C
ESD_HBM	ESD according Human Body Model (HBM) Q100-002	-2000		2000	V
ESD_CDM	ESD according Charged Device Model (CDM) Q100-011	-1000		1000	V
US	Ultrasonic agitation (cleaning, welding, etc.)	Prohibited			

2.7 AEC-Q100 Testing

The SCL3300 product family is tested according to AEC-Q100 Grade 1, revision H. Deviations to the requirements are presented in Table 5.

Table 5. Deviations to AEC-Q100 requirements.

Stress	ABV	Requirement	Deviation
Package Drop	DROP	Drop part on each of 6 axes once from a height of **1.2m** onto a concrete surface.	10 Drops, random orientation, drop height **0.8m**[1]

[1] Shocks may cause mechanical damage to the internal structures of MEMS sensor, causing malfunction of sensor, therefore mechanical shocks should be avoided. The level depends heavily on the pulse width and shape and should be evaluated case by case. As a general guideline, the lighter assembly or part, the higher shock levels will be generated on sensor component. **Dropped components shall not be used and shall be scrapped.**

2.6 绝对最大额定值

在最大额定值（表 4）内，组件不应发生损坏。参数值可能偏离规格，但不应发生功能失效。

表 4. 绝对最大额定值。

符号	描述	Min.	Typ	Max.	Unit
VDD	供电电压模拟电路	-0.3		4.3	V
DIN/DOUT	数字输入和输出引脚的最大电压	-0.3		DVIO+0.3	V
Topr	工作温度范围	-40		+125	°C
Tstg	存储温度范围	-40		+150	°C
ESD HBM_	ESD 根据人体模型（HBM）Q100-002	-2000		2000	V
ESD CDM_	根据带电设备模型（CDM）的静电放电 Q100-011	-1000		1000	V
US	超声波搅拌（清洗、焊接等）	禁止			

2.7 AEC-Q100 测试

SCL3300 产品系列按照 AEC-Q100 等级 1、修订版 H 进行测试。对要求的偏差详见表 5。

表 5. 对 AEC-Q100 要求的偏差。

压力	ABV	需求	偏差
包下载	DROP	在 6 个轴中的每一个上，从 1.2 米的高	10 滴，随机方向，下落高度 0.8 米[1]

[1] 冲击可能导致 MEMS 传感器内部结构发生机械损伤，从而引起传感器故障，因此应避免机械冲击。冲击等级高度依赖于脉冲宽度和波形，需根据具体情况评估。作为一般指导原则，组件或部件越轻，传感器组件上产生的冲击水平越高。跌落组件不得使用，应报废处理。

2.8 Pin Description

The pinout for SCL3300-D01 is presented in Figure 1.

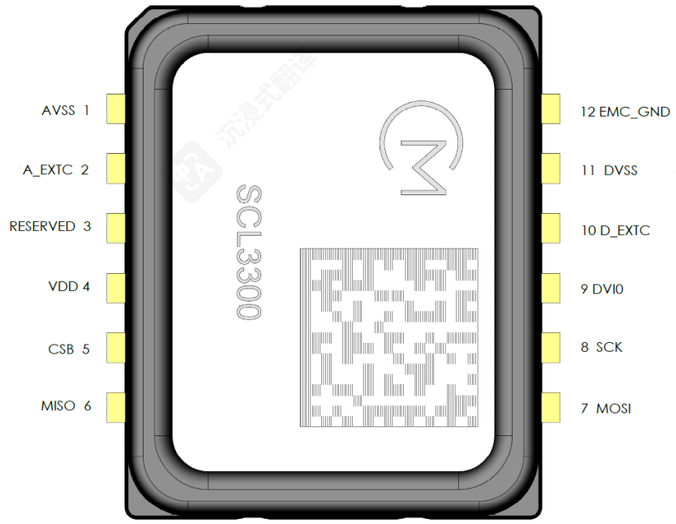


Figure 1 Pinout for SCL3300-D01.

Table 6 SCL3300-D01 pin descriptions.

Pin#	Name	Type	Description
1	AVSS	GND	Analog Reference Ground, connect externally to GND
2	A_EXTC	AOUT	External capacitor connection for analog core
3	RESERVED	-	Factory use only, connect externally to GND
4	VDD	SUPPLY	Analog Supply Voltage
5	CSB	DIN	Chip Select of SPI Interface, 3.3V logic compatible Schmitt-trigger input
6	MISO	DOUT	Data Out of SPI Interface
7	MOSI	DIN	Data In of SPI Interface, 3.3V logic compatible Schmitt-trigger input
8	SCK	DIN	CLK Signal of SPI Interface, 3.3V logic compatible Schmitt-trigger input
9	DVIO	SUPPLY	SPI Interface Supply Voltage. Must never be higher than VDD
10	D_EXTC	AOUT	External capacitor connection for digital core
11	DVSS	GND	Digital Reference Ground, connect externally to GND. Must never be left floating when component is powered.
12	EMC_GND	EMC GND	EMC Ground, connect externally to GND

2.8 引脚说明

SCL3300-D01 的引脚定义如图 1 所示。

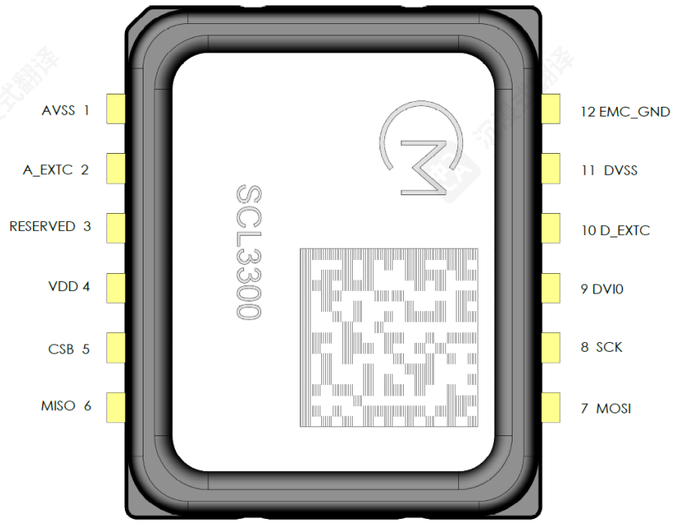


图 1 SCL3300-D01 引脚定义。

表 6 SCL3300-D01 引脚说明。

Pin#	Name	Type	描述
1	AVSS	GND	模拟参考地，外部连接至 GND
2	A EXTC_	AOUT	模拟核心外部电容连接
3	保留	-	仅限工厂使用，外部连接至 GND
4	VDD	供电模拟电	
5	CSB	DIN	SPI 接口片选，3.3V 逻辑兼容施密特触发器输入
6	MISO	DOUT	SPI 接口数据输出
7	MOSI	DIN	SPI 接口数据输入，兼容 3.3V 逻辑的施密特触发器输入
8	SCK	DIN	SPI 接口时钟信号，3.3V 逻辑兼容施密特触发器输入
9	DVIO	供应	SPI 接口供电电压。必须始终不超过 VDD。
10	D EXTC_	AOUT	数字核心外部电容连接
11	DVSS	GND	数字参考地，需外接至 GND。当器件通电时，绝不可悬空。
12	EMC GND_	EMC GND	

2.9 Typical performance characteristics

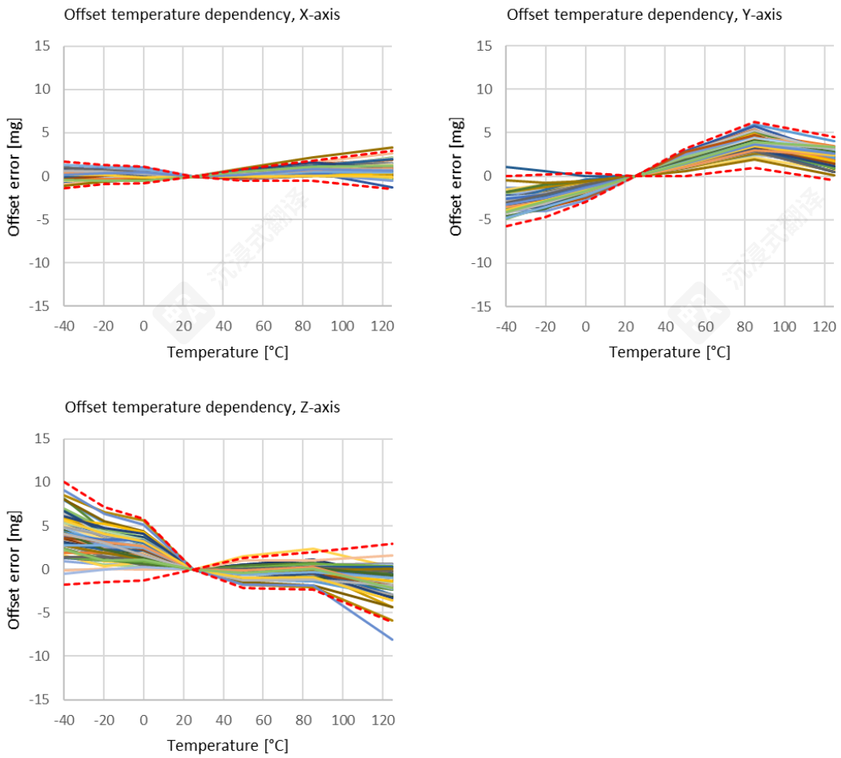


Figure 2 Accelerometer typical offset temperature behavior. Dotted lines show the ±3σ variation of the population.

2.9 典型性能特征

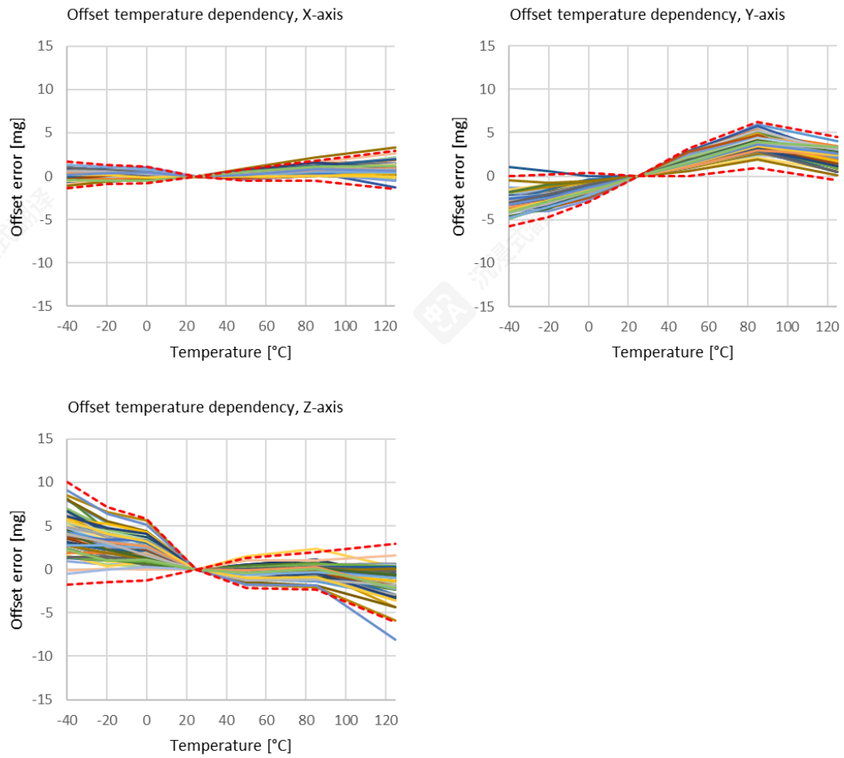


图 2 加速度计典型偏移温度特性。虚线显示 ±3σ 人群的变化。

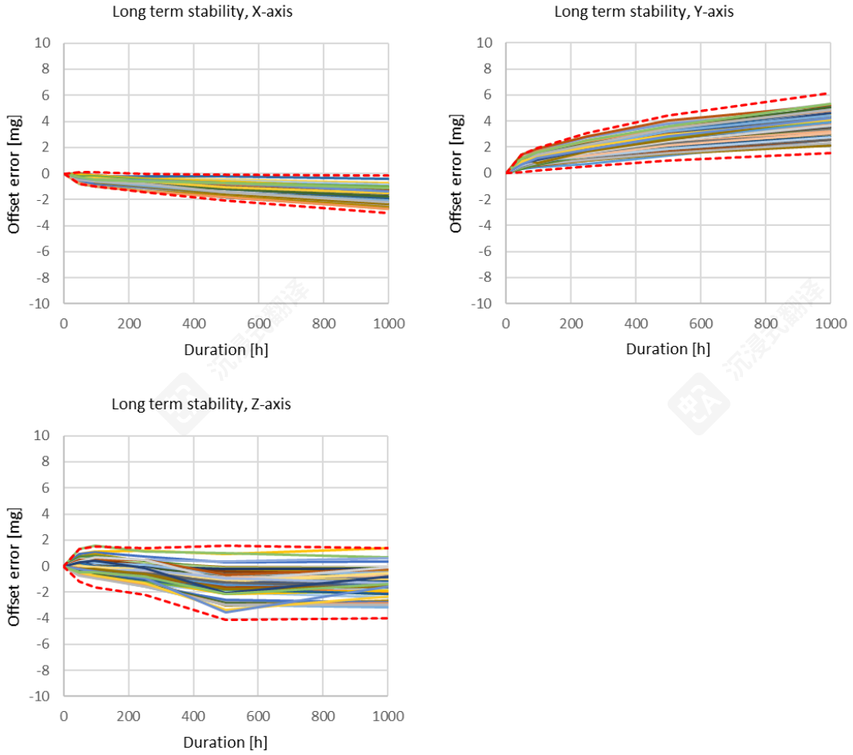


Figure 3 Example of accelerometer long term stability during 1000h HTOL. Test condition = +85 °C, Vsupply=3.6 V. Data measurement condition = +25 °C. Dotted lines show the ±3σ variation of the population.

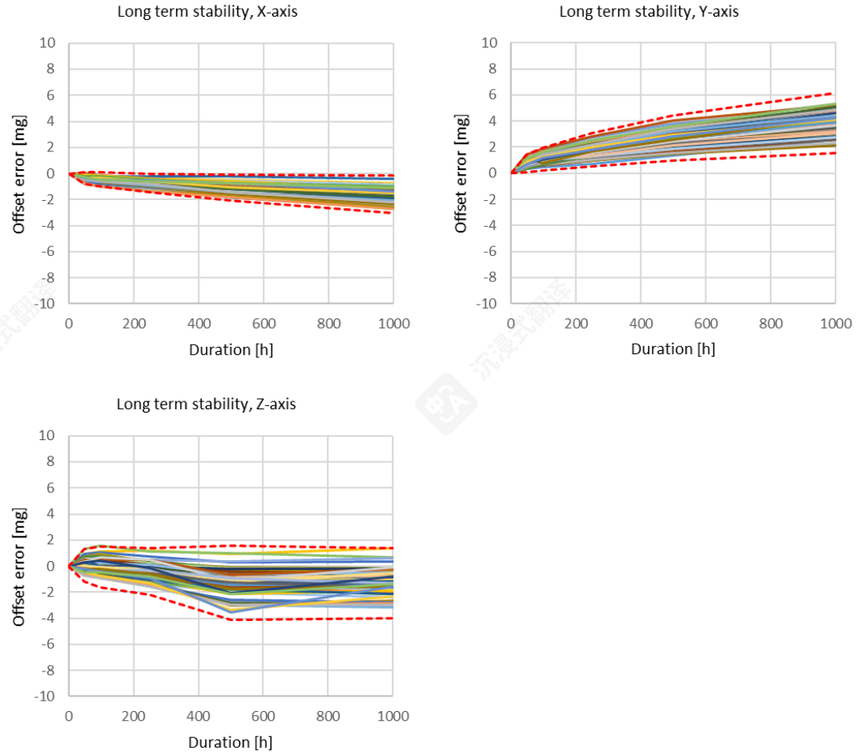


图 3 1000 = 加速度计在 +85 =3 6 = +25 ±3σ h HTOL 期间的长期稳定性示例。测试条件：°C，Vsupply . V。数据测量条件：°C。虚线显示总体变化的波动。

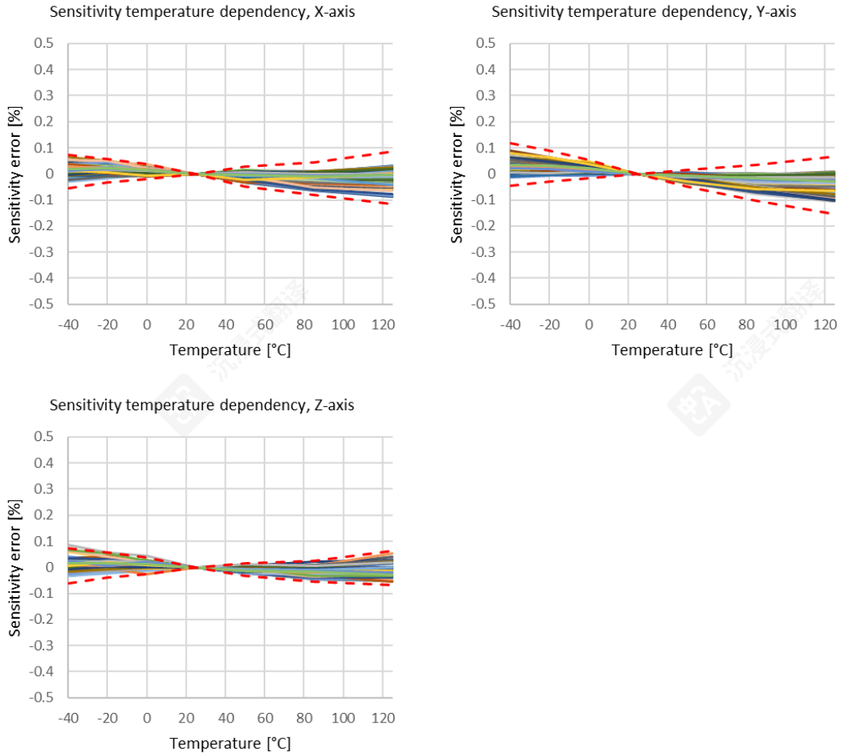


Figure 4 Accelerometer typical sensitivity temperature error in %

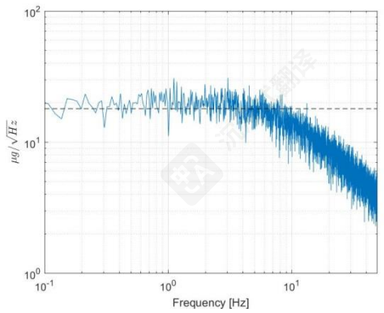


Figure 5 Example noise spectrum of X-channel in mode 4

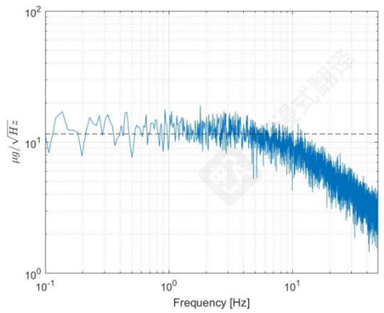


Figure 6 Example noise spectrum of Y-channel in mode 4

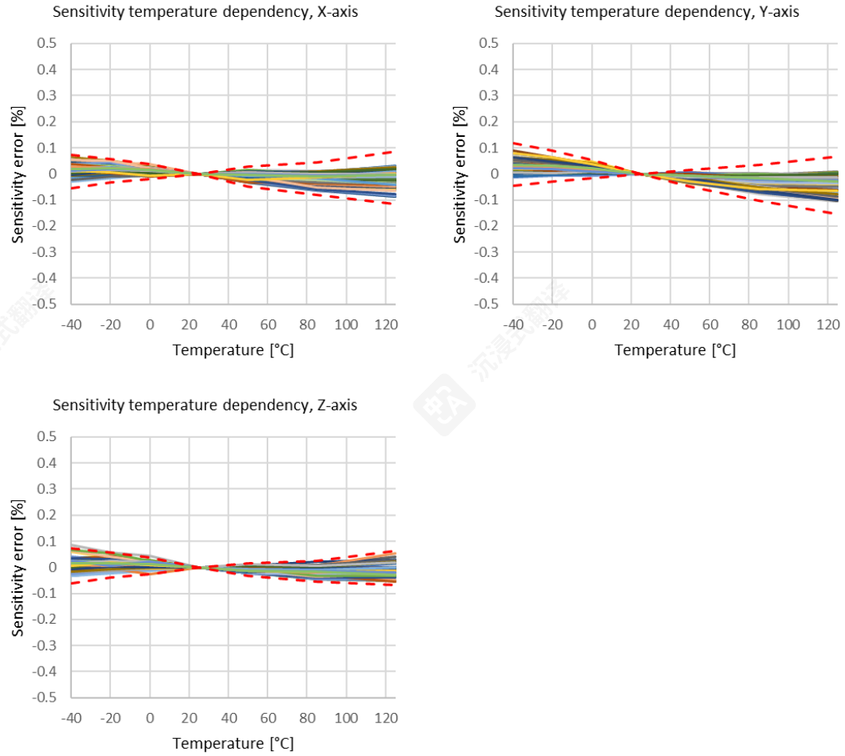


图 4 加速度计典型灵敏度温度误差（%）

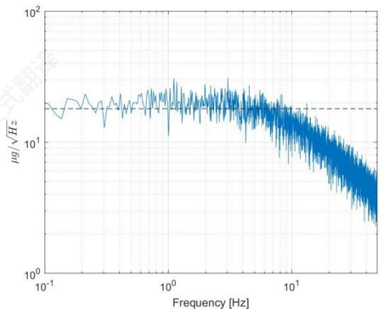


图 5 模式 4 下 X 通道的示例噪声频谱

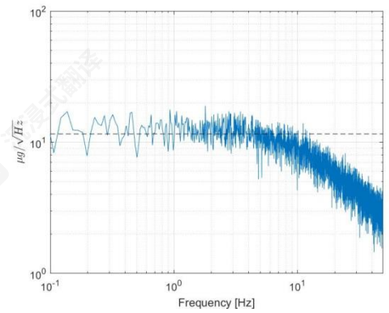


图 6 模式 4 中 Y 通道的示例噪声频谱

2.10 Digital I/O Specification

2.10.1 SPI DC Characteristics

Table 7 describes the DC characteristics of SCL3300-D01 sensor SPI I/O pins. Supply voltage is 3.3 V unless otherwise specified. Current flowing into the circuit has a positive value.

Table 7 SPI DC Characteristics

Symbol	Remark		Min.	Typ	Max.	Unit
Serial Clock SCK (Pull Down)						
I_{PD}	Pull-down current	Vin = 3.0 - 3.6 V	7.5	16.5	36	uA
V_{IH}	Input voltage '1'		0.67*DVIO		DVIO	V
V_{IL}	Input voltage '0'		0		0.33*DVIO	V
Chip Select CSB (Pull Up), low active						
I_{PU}	Pull-up current	Vin = 0	7.5	16.5	36	uA
V_{IH}	Input voltage '1'		0.67*DVIO		DVIO	V
V_{IL}	Input voltage '0'		0		0.33*DVIO	V
Serial Data Input MOSI (Pull Down)						
I_{PD}	Pull-down current	Vin = 3.0 - 3.6 V	7.5	16.5	36	uA
V_{IH}	Input voltage '1'		0.67*DVIO		DVIO	V
V_{IL}	Input voltage '0'		0		0.33*DVIO	V
Serial Data Output MISO (Tri State)						
V_{OH}	Output high voltage	I > -1 mA	DVIO-0.5V			V
V_{OL}	Output low voltage	I < 1 mA			0.5	V
I_{LEAK}	Tri-state leakage	0 < VMISO < 3.3 V	-1	0	1	uA
	Maximum Capacitive load				50	pF

2.10 数字 I/O 规范

2.10.1 SPI 直流特性

表 7 描述了 SCL3300-D01 传感器 SPI I/O 引脚的直流特性。除非另有说明，供电电压为 3.3 V。流入电路的电流为正值。

表 7 SPI 直流特性

符号	备注		Min.	Typ	Max.	Unit
串行时钟 SCK（下拉）						
I_{PD}	下拉电流	Vin = 3.0 - 3.6 V	7.5	16.5	36	uA
V_{IH}	输入电压'1'		0.67*DVIO		DVIO	V
V_{IL}	输入电压'0'		0		0.33*DVIO	V
片选 CSB（上拉），低电平有效						
I_{PU}	上拉电流	Vin = 0	7.5	16.5	36	uA
V_{IH}	输入电压'1'		0.67*DVIO		DVIO	V
V_{IL}	输入电压'0'		0		0.33*DVIO	V
串行数据输入 MOSI（下拉）						
I_{PD}	下拉电流	Vin = 3.0 - 3.6 V	7.5	16.5	36	uA
V_{IH}	输入电压'1'		0.67*DVIO		DVIO	V
V_{IL}	输入电压'0'		0		0.33*DVIO	V
串行数据输出 MISO（三态）						
V_{OH}	输出高压	我 > -1 mA	DVIO-0.5V			V
V_{OL}	输出低压	我 < 1 mA			0.5	V
我LEAK	三态泄漏	0 < VMISO < 3.3 V	-1	0	1	uA
	最大电容负载				50	pF

2.10.2 SPI AC Characteristics

The AC characteristics of SCL3300-D01 are defined in Figure 7 and Table 8.

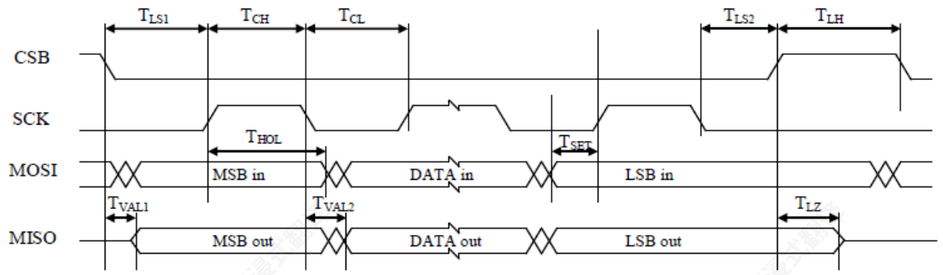


Figure 7 Timing diagram of SPI communication.

Table 8 SPI AC electrical characteristics.

Symbol	Description	Min.	Typ	Max.	Unit
T_{LS1}	Time from CSB (10%) to SCK (90%)	$T_{per}/2$			ns
T_{LS2}	Time from SCK (10%) to CSB (90%)	$T_{per}/2$			ns
T_{CL}	SCK low time	$T_{per}/2$			ns
T_{CH}	SCK high time	$T_{per}/2$			ns
$f_{SCK} = 1/T_{per}$	SCK Frequency *	0.1	2	8	MHz
T_{SET}	Time from changing MOSI (10%, 90%) to SCK (90%). Data setup time	$T_{per}/4$			ns
T_{HOL}	Time from SCK (90%) to changing MOSI (10%, 90%). Data hold time	$T_{per}/4$			ns
T_{VAL1}	Time from CSB (10%) to stable MISO (10%, 90%)		10		ns
T_{LZ}	Time from CSB (90%) to high impedance state of MISO		10		ns
T_{VAL2}	Time from SCK (10%) to stable MISO (10%, 90%)		10		ns
T_{LH}	**Time between SPI cycles, CSB at high level (90%)**	**10**			**us**

* SPI communication may affect the noise level. Used SPI clock should be carefully validated. Recommended SPI clock is 2 MHz - 4 MHz to achieve the best performance.

2.10.2 SPI AC 特性

SCL3300-D01 的 AC 特性定义于图 7 和表 8。

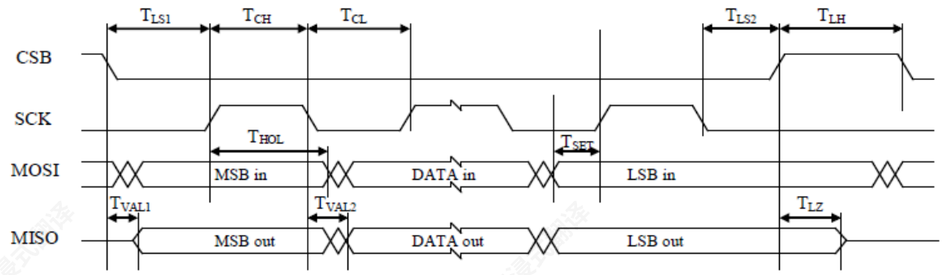


图 7 SPI 通信时序图。

表 8 SPI AC 电气特性。

符号	描述	Min.	Typ	Max.	Unit
T_{LS1}	从 CSB（10%）到 SCK（90%）的时间	$T_{per}/2$			ns
T_{LS2}	从 SCK (10%) 到 CSB (90%) 的时间	$T_{per}/2$			ns
T_{CL}	SCK 低电平时间	$T_{per}/2$			ns
T_{CH}	SCK 高时间	$T_{per}/2$			ns
$f_{SCK} = 1/T_{per}$		0.1	2	8	MHz
T_{SET}	从改变 MOSI（10%，90%）到 SCK（90%）的时间。	$T_{per}/4$			ns
T_{HOL}	从 SCK（90%）到改变 MOSI（10%，90%）的时间。	$T_{per}/4$			ns
T_{VAL1}	从 CSB（10%）到稳定 MISO（10%，90%）的时间		10		ns
T_{LZ}	CSB（90%）到 MISO 高阻抗状态的时间		10		ns
T_{VAL2}	从 SCK (10%) 到稳定 MISO (10%, 90%) 的时间		10		ns
T_{LH}	SPI 周期之间的时间，CSB 处于高电平（90%）	**10**			**us**

* SPI 通信可能会影响噪声水平。应仔细验证使用的 SPI 时钟频率。为实现最佳性能，推荐的 SPI 时钟频率为 2 MHz 至 4 MHz。

2.11 Measurement Axis and Directions

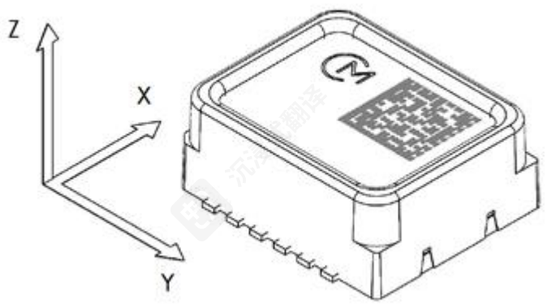


Figure 8 SCL3300-D01 measurement directions.

Table 9 SCL3300-D01 accelerometer measurement directions.

		
x: +1g angle x: 90° y: 0g angle y: 0° z: 0g angle z: 0°	x: 0g angle x: 0° y: +1g angle y: 90° z: 0g angle z: 0°	x: 0g angle x: 0° y: 0g angle y: 0° z: +1g angle z: 90°
		
x: -1g angle x: 270° y: 0g angle y: 0° z: 0g angle z: 0°	x: 0g angle x: 0° y: -1g angle y: 270° z: 0g angle z: 0°	x: 0g angle x: 0° y: 0g angle y: 0° z: -1g angle z: 270°

2.11 测量轴与方向

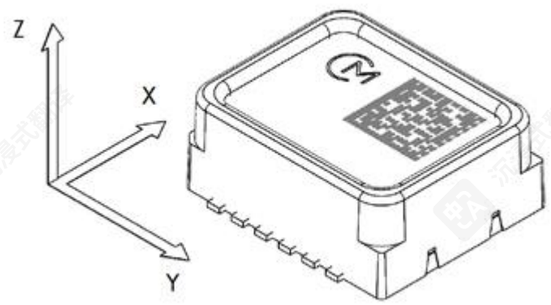


图 8 SCL3300-D01 测量方向。

表 9 SCL3300-D01 加速度计测量方向。

		
x: +1g 角度 x: 90° y: 0g 角度 y: 0° z: 0g 角度 z: 0°	x: 0g 角度 x: 0° y: +1g 角度 y: 90° z: 0g 角度 z: 0°	x: 0g 角度 x: 0° y: 0g 角度 y: 0° z: +1g 角度 z: 90°
		
x: -1g 角度 x: 270° y: 0g 角度 y: 0° z: 0g 角度 z: 0°	x: 0g 角度 x: 0° y: -1g 角度 y: 270° z: 0g 角度 z: 0°	x: 0g 角度 x: 0° y: 0g 角度 y: 0° z: -1g 角度 z: 270°

2.11.1 Measurement Ranges on Inclination Modes

Inclination ranges are limited in Mode 3 and Mode 4 to maximum ±10° inclination. See Figure 9 Measurement ranges on inclination modes (Mode 3 and Mode 4) below. If the whole 360° operation is needed, then one should select either Mode 1 or Mode 2 where the limitations regarding the maximum inclination angle don't exist.

The orientation in which the Y-axis is parallel to gravity (i.e. ±1g) is not recommended when using either mode 3 or mode 4. The dynamic range in that direction is limited and it is possible that saturation flag[2] may give an alert even if the tilt angle is less than 10°. Performance specifications according to Table 2 are met.

SCL3300-D01 inclination measurement is based on measuring the angles between the component and the gravity vector in static environment. Note that no other accelerations should be present in order to measure angles correctly.

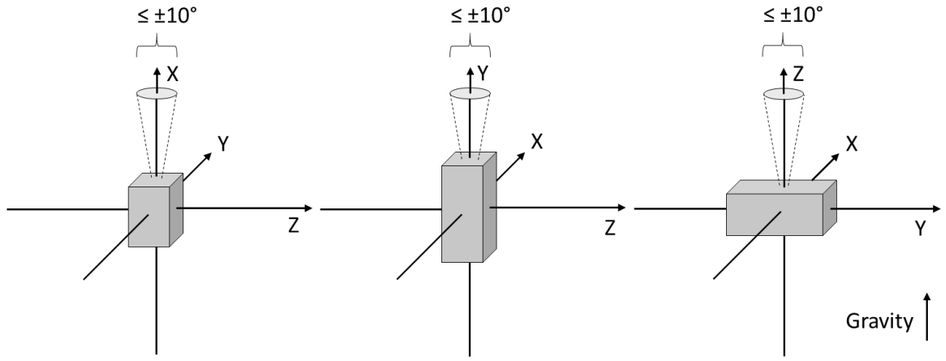


Figure 9 Measurement ranges on inclination modes (Mode 3 and Mode 4)

[2] See Chapter 6.3 for more information

2.11.1 倾斜模式下的测量范围

Mode 3 和 Mode 4 的倾角范围限制在最大 ±10° 倾角。详见下文图 9“倾角模式（Mode 3 和 Mode 4）的测量范围”。如果需要整个 360° 操作，则应选择 Mode 1 或 Mode 2，其中不存在关于最大倾角角度的限制。

当 Y 轴与重力方向平行（即 ±1g）时，不建议使用模式 3 或模式 4。该方向上的动态范围有限，即使倾斜角小于 10°，饱和标志[2] 也可能发出警报。表 2 中的性能规格已满足要求。

SCL3300-D01 倾角测量基于在静态环境中测量组件与重力矢量之间的角度。请注意，为了正确测量角度，不应存在其他加速度。

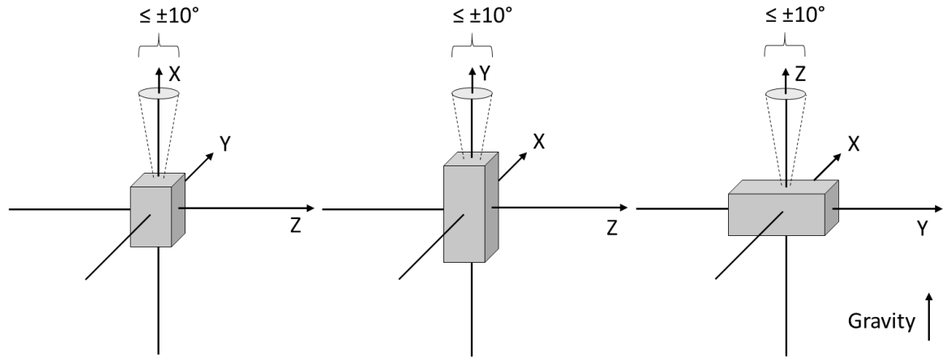


图 9 倾斜模式（模式 3 和模式 4）的测量范围

[2] 更多信息请参阅第 6.3 章

2.12 Package Characteristics

2.12.1 Package Outline Drawing

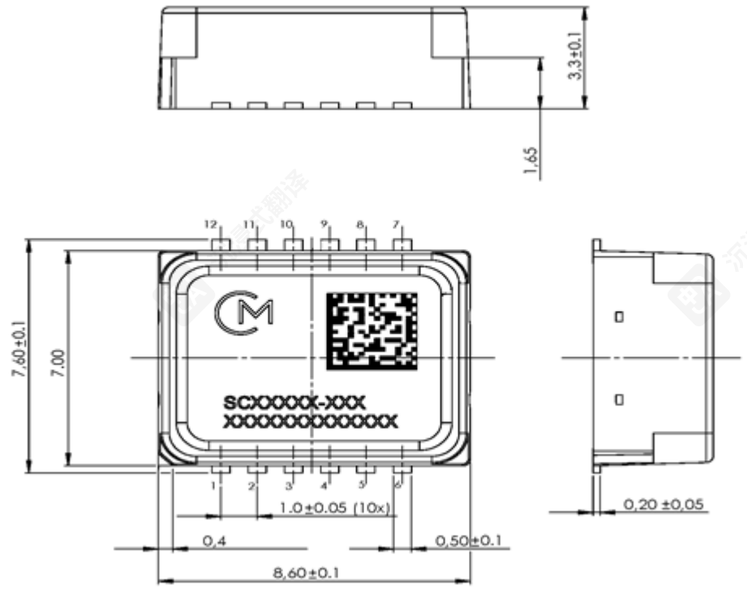


Figure 10 Package outline. The tolerances are according to ISO2768-f (see Table 10).

Table 10 Limits for linear measures (ISO2768-f).

Tolerance class	Limits in mm for nominal size in mm		
	0.5 to 3	Above 3 to 6	Above 6 to 30
f (fine)	±0.05	±0.05	±0.1

2.12 包特性

2.12.1 包外形图

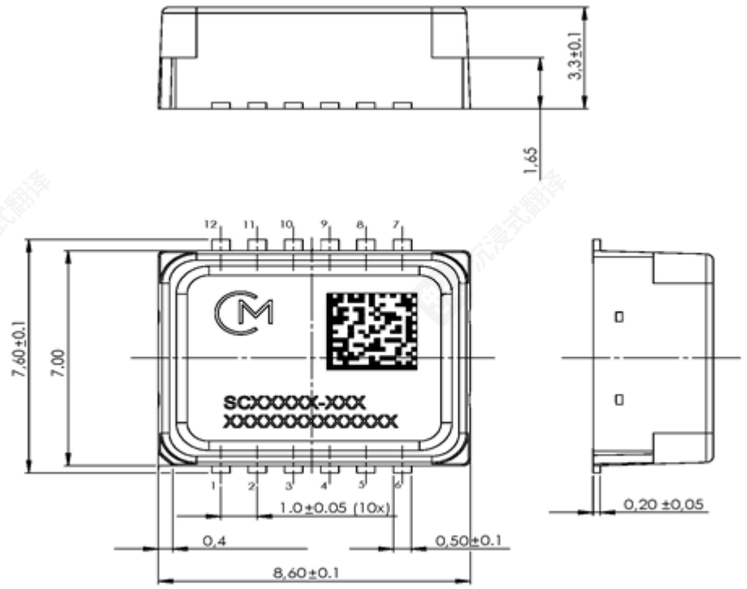


图 10 封装外形。公差依据 ISO2768-f（见表 10）。

表 10 线性尺寸的公差（ISO2768-f）。

公差等级	公称尺寸（mm）对应的极限尺寸（mm）		
	0.5 到 3	3 到 6 以上	6 到 30 以上
f (良好)	±0.05	±0.05	±0.1

2.13 PCB Footprint

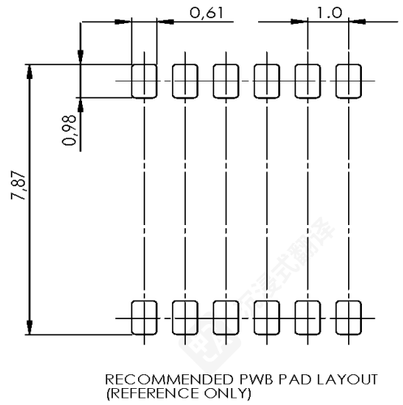


Figure 11 Recommended PWB pad layout for SCL3300-D01. All dimensions are in mm. The tolerances are according to ISO2768-f (see Table 10).

2.13 PCB 焊盘

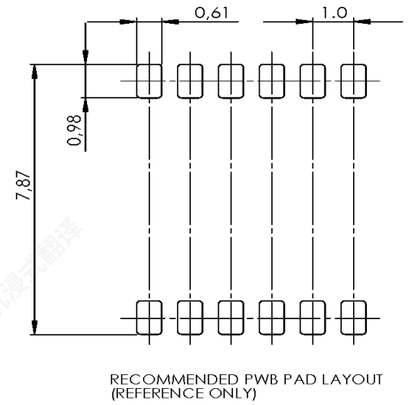


图 11 SCL3300-D01 推荐 PWB 焊盘布局。所有尺寸单位为毫米。公差符合 ISO2768-f（见表 10）。

3 General Product Description

The SCL3300-D01 sensor includes acceleration sensing element and Application-Specific Integrated Circuit (ASIC). Figure 12 contains an upper level block diagram of the component.

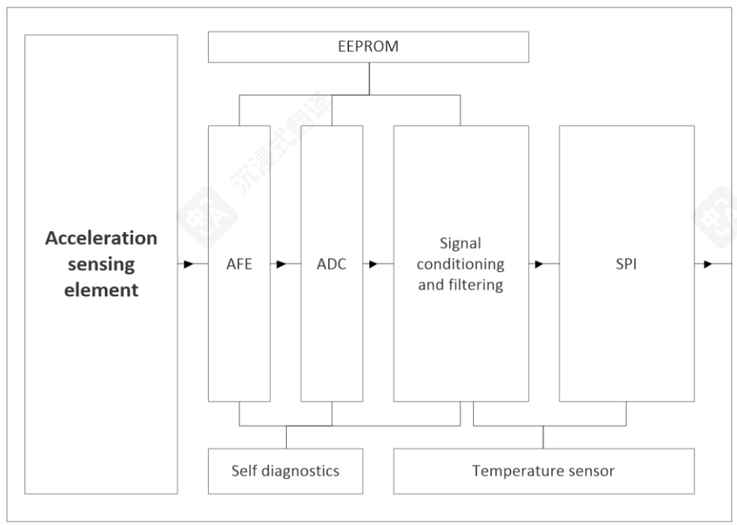


Figure 12. SCL3300-D01 component block diagram.

The sensing elements are manufactured using Murata proprietary High Aspect Ratio (HAR) 3D-MEMS process, which enables making robust, extremely stable and low noise capacitive sensors.

The acceleration sensing element consists of four acceleration sensitive masses. Acceleration causes capacitance change that is converted into a voltage change in the signal conditioning ASIC.

3.1 Factory Calibration

SCL3300-D01 sensors are factory calibrated. No separate calibration is required in the application. Calibration parameters are stored to non-volatile memory during manufacturing. The parameters are read automatically from the internal non-volatile memory during the start-up.

Assembly can cause offset/bias errors to the sensor output. If best possible accuracy is required, system level offset/bias calibration (zeroing) after assembly is recommended. Offset calibration is recommended to be performed not earlier than 12 hours after reflow. It should be noted that accuracy can be improved with longer stabilization time.

3 通用产品描述

SCL3300-D01 传感器包含加速度传感元件和专用集成电路（ASIC）。图 12 包含该组件的顶层框图。

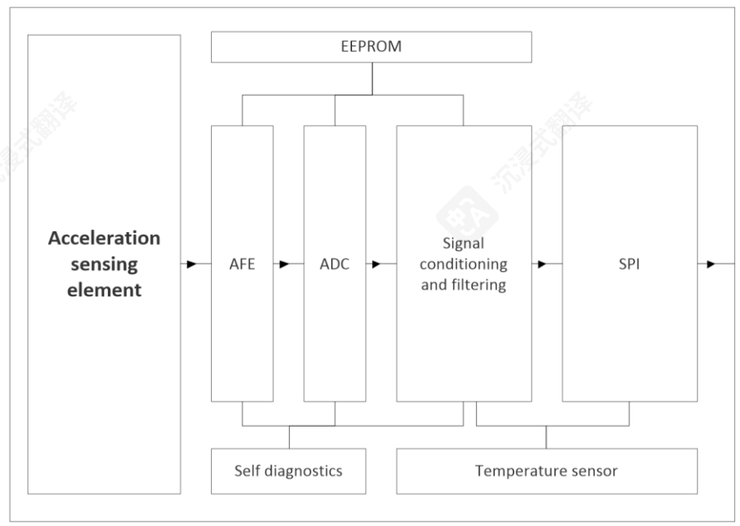


图 12。SCL3300-D01 组件框图。

传感元件采用村田公司专有的高长径比（HAR）3D-MEMS 工艺制造，该工艺可制造出坚固、极其稳定且低噪声的电容式传感器。

加速度传感元件由四个加速度敏感质量块组成。加速度引起的电容变化被转换为信号调理 ASIC 中的电压变化。

3.1 工厂校准

SCL3300-D01 传感器出厂已校准，应用时无需单独校准。校准参数在制造过程中已存储至非易失性存储器中。参数将从内部非易失性存储器自动读取。

启动期间的内存。
组装可能导致传感器输出出现偏移/偏差误差。若需要最佳精度，建议在组装后进行系统级偏移/偏差校准（归零）。推荐在回流后至少 12 小时进行偏移校准。需要注意的是，延长稳定时间可提高精度。

4 Component Operation, Reset and Power Up

4.1 Component Operation

Sensor ODR in normal operation mode is 2000 Hz. Registers are updated in every 0.5 ms and if all data is not read the full noise performance of sensor is not met.

In order to achieve optimal performance, it is recommended that during normal operation acceleration outputs ACCX, ACCY, ACCZ are read in every cycle using sensor ODR. It is necessary to read STATUS register only if return status (RS) indicates error.

4 组件操作、复位与上电

4.1 组件操作

正常操作模式下，传感器 ODR 为 2000 Hz。寄存器每 0.5 ms 更新一次，若未读取全部数据，则无法达到传感器的完整噪声性能。

为实现最佳性能，建议在正常运行期间，每个周期都使用传感器 ODR 读取加速度输出 ACCX、ACCY、ACCZ。仅在返回状态（RS）指示错误时才需要读取 STATUS 寄存器。

4.2 Start-up sequence

Table 11 Start-Up Sequence

Step	Procedure	RS*	Function	Note
1	Set VDD 3.0 - 3.6 V DVIO 3.0 - 3.6 V	- -		**Procedure for normal startup** VDD and DVIO don't need to rise at the same time
			OR	
1	Write Wake up from power down mode command	- -		**Procedure if device is in power down mode** See Table 15 for more information
1.2	Wait 1 ms	- -	Memory reading Settling of signal path	Only needed after power down mode
			Always continue from here	
2	Write SW Reset command	- -	Software reset the device	See Table 15 Operations and their equivalent SPI frames
3	Wait 1 ms	- -	Memory reading Settling of signal path	
4	Set Measurement mode**	'11'	Select operation mode	Mode 1 (default): 1.2g full-scale; 40 Hz 1st order low pass filter Mode 2: 2.4g full-scale; 70 Hz 1st order low pass filter Mode 3: Inclination mode; 10 Hz 1st order low pass filter Mode 4: Inclination mode; 10 Hz 1st order low pass filter; Low noise mode
5	Write ANG_CTRL	'01'	Enable angle outputs	See section 6.6 for more information.
6	Wait 25 ms	- -	Settling of signal path, Mode 1	
	OR Wait 15 ms	- -	Settling of signal path, Mode 2	
	OR Wait 100 ms	- -	Settling of signal path, Modes 3 and 4	
7	Read STATUS	'11'	Clear status summary	Reset status summary
8	Read STATUS	'11'	Read status summary	SPI response to step 5 Read status summary. Due to SPI off-frame protocol response is before STATUS has been cleared.
9	Read STATUS (or any other valid SPI command)	'01'	Ensure successful start-up	SPI response to step 6. First response where STATUS has been cleared. RS bits should be '01' to indicate proper start-up. Otherwise start-up has not been done correctly. See 6.3 STATUS for more information.

4.2 启动序列

表 11 启动序列

步骤		RS*	函数	Note
1	Set VDD 3.0 -)	- -		正常启动流程 VDD 和 DVIO 不需要在同时
			OR	
1	写入从掉电模式唤醒	- -		设备处于关机模式时的处理流程，更多
1.2	等待 1 毫秒	- -	内存读取信号路径稳定中	电源关闭模式后必需
			始终从此处继续	
2	写入 SW 复位命令	- -	软件重置设备	参见表 15 操作及其等效 SPI 帧
3	等待 1 毫秒	- -	内存读取 信号路径的收敛	
4	设置测量模式**	'11'	选择操作模式	模式 1（默认）：1.2g 全量程 40 Hz 模式 2：2.4g 全量程 70 模式 3：倾斜模式 10 Hz 一 模式 4：倾斜模式 10 Hz 一
5	输入 ANG CTRL_	'01'	启用角度输出	有关更多信息，请参阅第 6.6 节。
6	等待 25 毫秒	- -	信号路径的收敛，模式 1	
	OR 等待 15 毫秒	- -	信号路径的收敛，模式 2	
	或等待 100 毫秒	- -	信号路径的收敛，模式 3 和模	
7	查看状态	'11'	清除状态摘要	重置状态摘要
8	查看状态	'11'	查看状态摘要	SPI 对步骤 5 的响应 读取状态摘要。由于 SPI 关闭帧协议响
9	读取 STATUS（或任	'01'	确保成功启动	SPI 对步骤 6 的响应。 首次响应中 STATUS 为已清除。RS 位应为"01"以表示正确的启

* RS bits in returned SPI response during normal start-up. See 5.1.5 Return Status for more information.
** if not set, mode1 is used.
Please refer to Table 15 Operations and their equivalent SPI frames for detailed command frames.

4.3 Operation modes

SCL3300-D01 provides four user selectable operation modes.

Table 12 Operation mode description

Mode	Acceleration output			Inclination output	Acceleration and Inclination output
	Full-scale	Sensitivity LSB/g	Sensitivity LSB/° *	Sensitivity LSB/°	1st order low pass filter
1	± 1.2 g	6000	105	182	40 Hz
2	± 2.4 g	3000	52	182	70 Hz
3	Inclination mode**	12000	209	182	10 Hz
4	Inclination mode**	12000	209	182	10 Hz

* Angle calculated using 1g * SIN(θ), where θ is the inclination angle relative to the 0g position. Due to characteristics of sine function sensitivity is inversely proportional to inclination angle. Reported values are valid only between 0° to ±1°.

** Inclination mode. Dynamic range is dependent on orientation in gravity.

5 Component Interfacing

5.1.1 General

SPI communication transfers data between the SPI master and registers of the SCL3300-D01 ASIC. The SCL3300-D01 always operates as a slave device in master-slave operation mode. 3-wire SPI connection is not supported.

Table 13 SPI interface pins

Pin	Pin Name	Communication		
CSB	Chip Select (active low)	MCU	→	SCL3300
SCK	Serial Clock	MCU	→	SCL3300
MOSI	Master Out Slave In	MCU	→	SCL3300
MISO	Master In Slave Out	SCL3300	→	MCU

* RS bits in returned SPI response during normal start-up. See 5.1.5 Return Status for more information.
** if not set, mode1 is used.
请参照表 15“操作及其等效 SPI 帧”以获取详细的命令帧。

4.3 操作模式

SCL3300-D01 提供四种用户可选的操作模式。

表 12 操作模式说明

Mode	加速度输出			倾角输出	加速度和倾角输出
	全尺寸	灵敏度	灵敏度	灵敏度	1st 低通滤波
1	± 1.2 克	6000	105	182	40 赫兹
2	± 2.4 克	3000	52	182	70 赫兹
3	倾角模式**	12000	209	182	10 赫兹
4	倾角模式**	12000	209	182	10 赫兹

* 角度使用 1g * SIN(θ) 计算，其中 θ 是相对于 0g 位置的倾角。由于正弦函数的特性，其灵敏度与倾角成反比。报告值仅在 0° 至 ±1° 之间有效。

** 倾斜模式。动态范围取决于重力方向。

5 组件接口

5.1.1 概述

SPI 通信在 SPI 主设备与 SCL3300-D01 ASIC 的寄存器之间传输数据。SCL3300-D01 始终在主机 - 从机操作模式下作为从机设备运行。不支持 3 线 SPI 连接。

表 13 SPI 接口引脚

Pin	引脚名称	通信		
CSB	片选（低电平有效）	MCU	→	SCL3300
SCK	串行时钟	MCU	→	SCL3300
MOSI	主输出 从输入	MCU	→	SCL3300
MISO	主从输出	SCL3300	→	MCU

5.1.2 Protocol

The SPI is a 32-bit 4-wire slave configured bus. Off-frame protocol is used so each transfer consists of two phases. A response to the request is sent within next request frame. The response concurrent to the request contains the data requested by the previous command. The first bit in a sequence is an MSB.

The SPI transmission is always started with the falling edge of chip select, CSB. The data bits are sampled at the rising edge of the SCK signal. The data is captured on the rising edge (MOSI line) of the SCK and it is propagated on the falling edge (MISO line) of the SCK. This equals to SPI Mode 0 (CPOL = 0 and CPHA = 0).

NOTE: For sensor operation, time between consecutive SPI requests (i.e. CSB high) must be at least 10 µs. If less than 10 µs is used, output data will be corrupted.

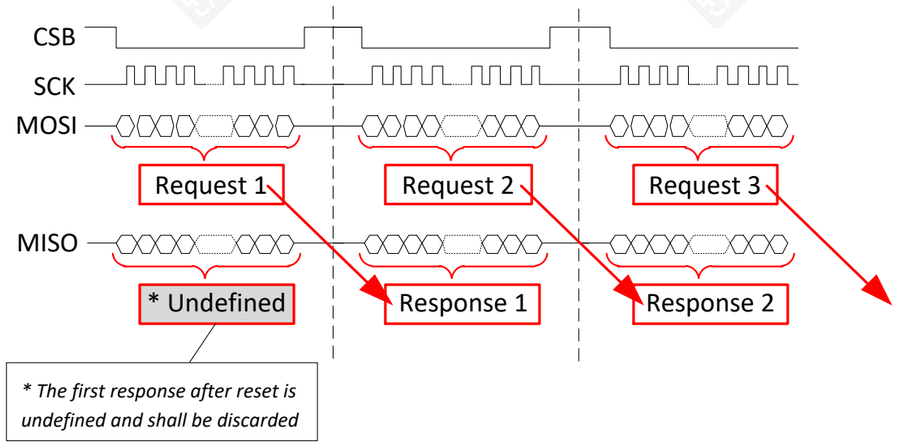


Figure 13 SPI Protocol

5.1.2 协议

SPI 是一种 32 位 4 线从机配置总线。采用帧外协议，因此每次传输由两个阶段组成。响应数据在下一个请求帧中发送。与请求并发传输的响应包含前一个命令所请求的数据。序列中的第一位是 MSB。

SPI 传输始终以片选 CSB 的下降沿开始。数据位采样于 SCK 信号的上升沿。数据在 SCK 的上升沿（MOSI 线）被捕获，并在 SCK 的下降沿（MISO 线）传播。这等同于 SPI 模式 0（CPOL = 0 和 CPHA = 0）。

注意：对于传感器操作，连续 SPI 请求（即 CSB 高电平）之间的时间间隔必须至少为 10 µs。若使用小于 10 µs 的时间间隔，输出数据将损坏。

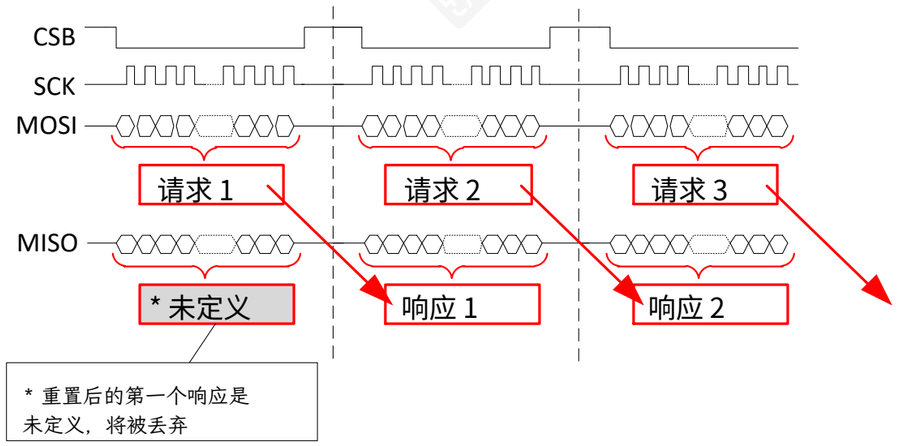


图 13 SPI 协议

5.1.3 SPI frame

The SPI Frame is divided into four parts:

1. Operation Code (OP), consisting of Read/Write (RW) and Address (ADDR)
2. Return Status (RS, in MISO)
3. Data (D)
4. Checksum (CRC)

See Figure 14 and Table 14Table 14 SPI Frame Specification for more details. For allowed SPI operating commands see Table 15.

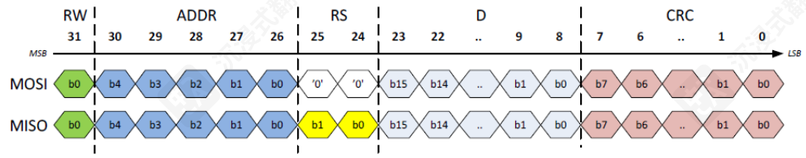


Figure 14 SPI Frame

Table 14 SPI Frame Specification

Name	Bits	Description	MISO / MOSI	
OP	[31:26]	Operation code RW + ADDR	OP [5] = RW OP [4:0] = ADDR	Read = 0 / Write = 1 Register address
RS	[25:24]	Return status	MISO '00' - Startup in progress '01' - Normal operation, no flags '10' - N/A '11' - Error	MOSI '00' – Always
D	[23:8]	Data	Returned data / data to write	
CRC	[7:0]	Checksum	See section 5.2	

Return Status (RS) shows error (i.e. '11') when an error flag (or flags) is active in, or if previous MOSI-command had incorrect CRC.

5.1.3 SPI 帧

SPI 帧分为四个部分：

1. 操作码（OP），由读写（RW）和地址（ADDR）组成
2. 返回状态（RS，在 MISO 中）
3. 数据 (D)
4. 校验和（CRC）

参见图 14 和表 14 表 14 SPI 帧规范以获取更多信息。有关允许的 SPI 操作命令，请参见表 15。

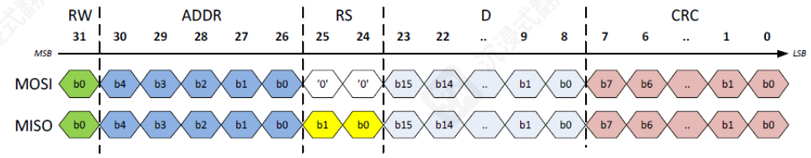


图 14 SPI 帧

表 14 SPI 帧规范

Name	Bits	描述	MISO / MOSI	
OP	[31:26]	操作码 RW +	OP [5] = RW OP [4:0] = ADDR	读取 = 0 / 写入 = 1 注册地址
RS	[25:24]	返回状态	MISO '00' - 启动中 '01' - 正常运行，无标志 '10' - 不	MOSI '00' – 始终
D	[23:8]		返回的数据 / 要写入的数据	
CRC	[7:0]	校验和	参见第 5.2 节	

返回状态（RS）在错误标志（或标志）激活时，或当之前的 MOSI 命令具有错误的 CRC 时，显示错误（即'11'）。

5.1.4 Operations

Allowed operation commands are shown in Table 15. No other commands are allowed.

Table 15 Operations and their equivalent SPI frames

Operation	Bank	SPI Frame	SPI Frame Hex
Read ACC_X	0 1	0000 0100 0000 0000 0000 0000 1111 0111	040000F7h
Read ACC_Y	0 1	0000 1000 0000 0000 0000 0000 1111 1101	080000FDh
Read ACC_Z	0 1	0000 1100 0000 0000 0000 0000 1111 1011	0C0000FBh
Read STO (self-test output)	0 1	0001 0000 0000 0000 0000 0000 1110 1001	100000E9h
Enable ANGLE outputs	0	1011 0000 0000 0000 0001 1111 0110 1111	B0001F6Fh
Read ANG_X	0	0010 0100 0000 0000 0000 0000 1100 0111	240000C7h
Read ANG_Y	0	0010 1000 0000 0000 0000 0000 1100 1101	280000CDh
Read ANG_Z	0	0010 1100 0000 0000 0000 0000 1100 1011	2C0000CBh
Read Temperature	0 1	0001 0100 0000 0000 0000 0000 1110 1111	140000EFh
Read Status Summary	0 1	0001 1000 0000 0000 0000 0000 1110 0101	180000E5h
Read ERR_FLAG1	0	0001 1100 0000 0000 0000 0000 1110 0011	1C0000E3
Read ERR_FLAG2	0	0010 0000 0000 0000 0000 0000 1100 0001	200000C1h
Read CMD	0	0011 0100 0000 0000 0000 0000 1101 1111	340000DFh
Change to mode 1	0	1011 0100 0000 0000 0000 0000 0001 1111	B400001Fh
Change to mode 2	0	1011 0100 0000 0000 0000 0001 0000 0010	B4000102h
Change to mode 3	0	1011 0100 0000 0000 0000 0010 0010 0101	B4000225h
Change to mode 4	0	1011 0100 0000 0000 0000 0011 0011 1000	B4000338h
Set power down mode	0	1011 0100 0000 0000 0000 0100 0110 1011	B400046Bh
Wake up from power down mode	0	1011 0100 0000 0000 0000 0000 0001 1111	B400001Fh
SW Reset	0	1011 0100 0000 0000 0010 0000 1001 1000	B4002098h
Read WHOAMI	0	0100 0000 0000 0000 0000 0000 1001 0001	40000091h
Read SERIAL1	1	0110 0100 0000 0000 0000 0000 1010 0111	640000A7h
Read SERIAL2	1	0110 1000 0000 0000 0000 0000 1010 1101	680000ADh
Read current bank	0 1	0111 1100 0000 0000 0000 0000 1011 0011	7C0000B3h
Switch to bank #0	0 1	1111 1100 0000 0000 0000 0000 0111 0011	FC000073h
Switch to bank #1	0 1	1111 1100 0000 0000 0000 0001 0110 1110	FC00016Eh

5.1.4 操作

允许的操作命令如表 15 所示。不允许其他命令。

表 15 操作及其等效 SPI 帧

操作	银行 S		SPI 帧十六进制
阅读 ACC _X	0 1	0000 0100 0000 0000 0000 0000 1111 0111	040000F7h
读取 ACC Y_	0 1	0000 1000 0000 0000 0000 0000 1111 1101	080000FDh
阅读 ACC Z_	0 1	0000 1100 0000 0000 0000 0000 1111 1011	0C0000FBh
读取 STO（自检输出）	0 1	0001 0000 0000 0000 0000 0000 1110 1001	100000E9h
启用 ANGLE 输出	0	1011 0000 0000 0000 0001 1111 0110 1111	B0001F6Fh
阅读 ANG _X	0	0010 0100 0000 0000 0000 0000 1100 0111	240000C7h
阅读 ANG Y_	0	0010 1000 0000 0000 0000 0000 1100 1101	280000CDh
阅读 ANG Z_	0	0010 1100 0000 0000 0000 0000 1100 1011	2C0000CBh
读取温度	0 1	0001 0100 0000 0000 0000 0000 1110 1111	140000EFh
读取状态摘要	0 1	0001 1000 0000 0000 0000 0000 1110 0101	180000E5h
阅读 ERR _FLAG1	0	0001 1100 0000 0000 0000 0000 1110 0011	1C0000E3
查看 ERR FLAG2_	0	0010 0000 0000 0000 0000 0000 1100 0001	200000C1h
读取 CMD	0	0011 0100 0000 0000 0000 0000 1101 1111	340000DFh
切换到模式 1	0	1011 0100 0000 0000 0000 0000 0001 1111	B400001Fh
切换到模式 2	0	1011 0100 0000 0000 0000 0001 0000 0010	B4000102h
切换到模式 3	0	1011 0100 0000 0000 0000 0010 0010 0101	B4000225h
切换到模式 4	0	1011 0100 0000 0000 0000 0011 0011 1000	B4000338h
设置关机模式	0	1011 0100 0000 0000 0000 0100 0110 1011	B400046Bh
从关机模式唤醒	0	1011 0100 0000 0000 0000 0000 0001 1111	B400001Fh
SW 复位	0	1011 0100 0000 0000 0010 0000 1001 1000	B4002098h
阅读 WHOAMI	0	0100 0000 0000 0000 0000 0000 1001 0001	40000091h
读取 SERIAL1	1	0110 0100 0000 0000 0000 0000 1010 0111	640000A7h
读取 SERIAL2	1	0110 1000 0000 0000 0000 0000 1010 1101	680000ADh
读取当前银行	0 1	0111 1100 0000 0000 0000 0000 1011 0011	7C0000B3h
切换到银行 #0	0 1	1111 1100 0000 0000 0000 0000 0111 0011	FC000073h
切换到银行 #1	0 1	1111 1100 0000 0000 0000 0001 0110 1110	FC00016Eh

5.1.5 Return Status

SPI frame Return Status bits (RS bits) indicate the functional status of the sensor. See Table 16 for RS definitions.

Table 16 Return Status definitions

RS [1]	RS [0]	Description
0	0	Startup in progress
0	1	Normal operation, no flags
1	0	Reserved
1	1	Error

The priority of the return status states is from high to low: 00 → 11 → 01

Return Status (RS) shows error (i.e. '11') when an error flag (or flags) is active in Status Summary register, or if previous MOSI-command had incorrect frame CRC. See Table 27 for description of the Status Summary register.

5.2 Checksum (CRC)

For SPI transmission error detection a Cyclic Redundancy Check (CRC) is implemented, for details see Table 17.

Table 17 SPI CRC definition

Parameter	Value
Name	CRC-8
Width	8 bit
Poly	1Dh (generator polynom: X8+X4+X3+X2+1)
Init	FFh (initialization value)
XOR out	FFh (inversion of CRC result)

The CRC value used in system level software has to be initialized with FFh to ensure a CRC failure in case of stuck-at-0 and stuck-at-1 error on the SPI bus. C-programming language example for CRC calculation is presented in Figure 15. It can be used as is in an appropriate programming context.

5.1.5 返回状态

SPI 帧返回状态位（RS 位）指示传感器的功能状态。RS 定义请参见表 16。

表 16 返回状态定义

RS [1]	RS [0]	描述
0	0	启动中
0	1	正常运行，无标志
1	0	保留
1	1	错误

返回状态优先级的顺序从高到低为：00 → 11 → 01

返回状态（RS）在状态摘要寄存器中活动了错误标志（或标志）时，或在之前的 MOSI 命令具有错误的帧 CRC 时，显示错误（即'11'）。有关状态摘要寄存器的描述，请参阅表 27。

5.2 校验和（CRC）

对于 SPI 传输错误检测，已实现循环冗余校验（CRC），详见表 17。

表 17 SPI CRC 定义

参数	值
Name	CRC-8
宽度	8 位
Poly	1Dh（生成多项式：X8+X4+X3+X2+1）
Init	FFh（初始化值）
异或	FFh（CRC 结果的取反）

系统级软件中使用的 CRC 值必须初始化为 FFh，以确保在 SPI 总线出现挂 0 或挂 1 错误时发生 CRC 校验失败。图 15 展示了 CRC 计算的 C 语言编程示例。它可直接在合适的编程上下文中使用。

```
// Calculate CRC for 24 MSB's of the 32 bit dword
// (8 LSB's are the CRC field and are not included in CRC calculation)

uint8_t CalculateCRC(uint32_t Data)
{
  uint8_t BitIndex;
  uint8_t BitValue;
  uint8_t CRC;

  CRC = 0xFF;
  for (BitIndex = 31; BitIndex > 7; BitIndex--)
  {
    BitValue = (uint8_t)((Data >> BitIndex) & 0x01);
    CRC = CRC8(BitValue, CRC);
  }
  CRC = (uint8_t)~CRC;
  return CRC;
}

static uint8_t CRC8(uint8_t BitValue, uint8_t CRC)
{
  uint8_t Temp;

  Temp = (uint8_t)(CRC & 0x80);
  if (BitValue == 0x01)
  {
    Temp ^= 0x80;
  }
  CRC <<= 1;
  if (Temp > 0)
  {
    CRC ^= 0x1D;
  }
  return CRC;
}
```

Figure 15 C-programming language example for CRC calculation

In case of wrong CRC in MOSI write/read, RS bits “11” are set in the next SPI response, STATUS register is not changed, and write command is discarded. If CRC in MISO SPI response is incorrect, communication failure occurred.

CRC calculation example:

```
Read ACC_X register (04h)
SPI [31:8] = 040000h → CRC = F7h
SPI [7:0]  = F7h
SPI frame  = 040000F7h
```

6 Register Definition

SCL3300-D01 contains two user switchable register banks. Default register bank is #0. One should have register bank #0 always active, unless data from bank #1 is required. After reading data from bank #1 is finished, one should switch back to bank #0 to ensure no accidental read / writes in unwanted registers. See 6.9 SELBANK for more information for selecting active register bank. Table 18 shows overview of register banks and register addresses.


```
// Calculate CRC for 24 MSB's of the 32 bit dword
// (8 LSB's are the CRC field and are not included in CRC calculation)

uint8_t CalculateCRC(uint32_t Data)
{
  uint8_t BitIndex;
  uint8_t BitValue;
  uint8_t CRC;

  CRC = 0xFF;
  for (BitIndex = 31; BitIndex > 7; BitIndex--)
  {
    BitValue = (uint8_t)((Data >> BitIndex) & 0x01);
    CRC = CRC8(BitValue, CRC);
  }
  CRC = (uint8_t)~CRC;
  return CRC;
}

static uint8_t CRC8(uint8_t BitValue, uint8_t CRC)
{
  uint8_t Temp;

  Temp = (uint8_t)(CRC & 0x80);
  if (BitValue == 0x01)
  {
    Temp ^= 0x80;
  }
  CRC <<= 1;
  if (Temp > 0)
  {
    CRC ^= 0x1D;
  }
  return CRC;
}
```


图 15 C 语言程序 CRC 计算示例

若 MOSI 写入/读取出现 CRC 错误，则在下一个 SPI 响应中将 RS 位设置为"11"，STATUS 寄存器保持不变，且写入命令被丢弃。若 MISO SPI 响应中的 CRC 不正确，则发生通信失败。

CRC 计算示例：


```
Read ACC_X register (04h)
SPI [31:8] = 040000h → CRC = F7h
SPI [7:0]  = F7h
SPI frame  = 040000F7h
```


6 寄存器定义

SCL3300-D01 包含两个用户可切换的寄存器组。默认寄存器组为#0。除非需要#1组的数据，否则应始终保持#0组处于活动状态。在完成从#1组读取数据后，应切换回#0组，以确保不会意外读取/写入不需要的寄存器。有关选择活动寄存器组的更多信息，请参阅 6.9 SELBANK。表 18 显示了寄存器组和寄存器地址的概览。

Table 18 Register address space overview

Addr (hex)	Read/ Write	Register Bank		Description
		#0	#1	
01h	R	ACC_X	ACC_X	X-axis acceleration output in 2's complement format
02h	R	ACC_Y	ACC_Y	Y-axis acceleration output in 2's complement format
03h	R	ACC_Z	ACC_Z	Z-axis acceleration output in 2's complement format
04h	R	STO	STO	Self-test output in 2's complement format
05h	R	TEMPERATURE	TEMPERATURE	Temperature sensor output in 2's complement format
06h	R	STATUS	STATUS	Status Summary combining ERR_FLAG1 and ERR_FLAG2
07h	R	ERR_FLAG1	reserved	Error flags group1
08h	R	ERR_FLAG2	reserved	Error flags group2
09h	-	ANG_X	reserved	X-axis angle output in 2's complement format
0Ah	-	ANG_Y	reserved	Y-axis angle output in 2's complement format
0Bh	-	ANG_Z	reserved	Z-axis angle output in 2's complement format
0Ch	-	ANG_CTRL	reserved	Enable angle outputs
0Dh	R / W	MODE	reserved	Sets operation mode, SW Reset and Power down mode
0Eh	-	reserved	reserved	-
0Fh	-	reserved	reserved	-
10h	R	WHOAMI	reserved	8-bit register for component identification
11h	-	reserved	reserved	-
12h	-	reserved	reserved	-
13h	-	reserved	reserved	-
14h	-	reserved	reserved	-
15h	-	reserved	reserved	-
16h	-	reserved	reserved	-
17h	-	reserved	reserved	-
18h	-	reserved	reserved	-
19h	R	reserved	SERIAL1	Component serial part 1
1Ah	R	reserved	SERIAL2	Component serial part 2
1Bh	-	reserved	Factory Use	-
1Ch	-	reserved	Factory Use	-
1Dh	-	reserved	Factory Use	-
1Eh	-	reserved	reserved	-
1Fh	R / W	SELBANK	SELBANK	Switch between active register banks

User should not access Reserved nor Factory Use registers. Power-cycle, reset and power down mode will reset all written settings.

表 18 寄存器地址空间概览

Addr (十六进制)	读取/ 编写	注册银行		描述
		#0	#1	
01h	R	ACC X_	ACC X_	X 轴加速度输出为 2 的补码格式
02h	R	ACC Y_	ACC Y_	2 的补码格式下的 Y 轴加速度输出
03h	R	ACC Z_	ACC Z_	2 的补码格式下的 Z 轴加速度输出
04h	R	STO	STO	2 的补码格式下的自检输出
05h	R	温度	TEMPERATURE 温	
06h	R	状态	状态	结合 ERR_FLAG1 和 ERR_FLAG2 的状态摘要
07h	R	ERR FLAG1_	保留	错误标志组 1
08h	R	ERR FLAG2_	保留	错误标志组 2
09h	-	ANG X_	保留	X 轴角度输出为 2 的补码格式
0Ah	-	ANG Y_	保留	Y 轴角度输出为 2 的补码格式
0Bh	-	ANG Z_	保留	Z 轴角度以 2 的补码格式输出
0Ch	-	ANG CTRL_	保留	启用角度输出
0Dh	R /	MODE	保留	设置操作模式、软件复位和掉电模式
0Eh	-	保留	保留	-
0Fh	-	保留	保留	-
10h	R	我是谁	保留	8 位寄存器用于组件识别
11h	-	保留	保留	-
12h	-	保留	保留	-
13h	-	保留	保留	-
14h	-	保留	保留	-
15h	-	保留	保留	-
16h	-	保留	保留	-
17h	-	保留	保留	-
18h	-	保留	保留	-
19h	R	保留	SERIAL1	组件序列部分 1
1Ah	R	保留	SERIAL2	组件序列部分 2
1Bh	-	保留	工厂专用	-
1Ch	-	保留	工厂专用	-
1Dh	-	保留	工厂专用	-
1Eh	-	保留	保留	-
1Fh	R /	SELBANK	SELBANK	在活跃寄存器组之间切换

用户不应访问保留或工厂使用寄存器。上电循环、复位和关机模式将重置所有已写入的设置。

6.1 Sensor Data Block

Table 19 Sensor data block description

Bank	Addr	Name	No. of bits	Read / Write	Description
0 1	01h	ACC_X	16	R	X-axis acceleration output in 2's complement format
0 1	02h	ACC_Y	16	R	Y-axis acceleration output in 2's complement format
0 1	03h	ACC_Z	16	R	Z-axis acceleration output in 2's complement format
0 1	05h	TEMPERATURE	16	R	Temperature sensor output in 2's complement format. See section 2.4 for conversion equation.
0	09h	ANG_X	16	R	X-axis angle output in 2's complement format See section 6.1.3 for conversion equation.
0	0Ah	ANG_Y	16	R	Y-axis angle output in 2's complement format See section 6.1.3 for conversion equation.
0	0Bh	ANG_Z	16	R	Z-axis angle output in 2's complement format See section 6.1.3 for conversion equation.

Table 20 Sensor data block operations

Operation	SPI Frame	SPI Frame Hex
Read ACC_X	0000 0100 0000 0000 0000 0000 1111 0111	040000F7h
Read ACC_Y	0000 1000 0000 0000 0000 0000 1111 1101	080000FDh
Read ACC_Z	0000 1100 0000 0000 0000 0000 1111 1011	0C0000FBh
Read Temperature	0001 0100 0000 0000 0000 0000 1110 1111	140000EFh
Read ANG_X	0010 0100 0000 0000 0000 0000 1100 0111	240000C7h
Read ANG_Y	0010 1000 0000 0000 0000 0000 1100 1101	280000CDh
Read ANG_Z	0010 1100 0000 0000 0000 0000 1100 1011	2C0000CBh

6.1 传感器数据块

表 19 传感器数据块描述

银行地			位数	读取 /	描述
0 1	01h A		16	R	X 轴加速度输出为 2 的补码格式
0 1	02h A		16	R	2 的补码格式下的 Y 轴加速度输出
0 1	03h A		16	R	2 的补码格式下的 Z 轴加速度输出
0 1	05 时		16	R	2 的补码格式的温度传感器输出。 请参阅第 2.4 节中的转换公式。
0	09h	ANG X_	16	R	X 轴角度输出为 2 的补码格式。转换公式见第
0	0Ah	ANG Y_	16	R	Y 轴角度输出为 2 的补码格式。转换公式请参阅第
0	0Bh	ANG Z_	16	R	Z 轴角度输出为 2 的补码格式。转换方程见第

表 20 传感器数据块操作

操作	SPI 帧	SPI 帧十六进制
阅读 ACC _X	0000 0100 0000 0000 0000 0000 1111 0111	040000F7h
阅读 ACC _Y	0000 1000 0000 0000 0000 0000 1111 1101	080000FDh
阅读 ACC _Z	0000 1100 0000 0000 0000 0000 1111 1011	0C0000FBh
读取温度	0001 0100 0000 0000 0000 0000 1110 1111	140000EFh
阅读 ANG X_	0010 0100 0000 0000 0000 0000 1100 0111	240000C7h
阅读 ANG Y_	0010 1000 0000 0000 0000 0000 1100 1101	280000CDh
阅读 ANG Z_	0010 1100 0000 0000 0000 0000 1100 1011	2C0000CBh

6.1.1 Example of Acceleration Data Conversion

For example, if ACC_X register read results: ACC_X = 0500DC1Ch, the register content is converted to acceleration rate as follows:

OP[31:26] + RS[25:24]		Data[23:8]				CRC[7:0]	
0	5	0	0	D	C	1	C

OP + RS

```
05h = 0000 0101b
      0000 01b          = OP code = Read ACC_X
      01b               = return status (RS bits) = no error

Data = ACC_X register content
      00DCh
            00DCh → 220d      = in 2's complement format
            Acceleration:
            = 220 LSB / sensitivity(mode1)
            = 220 LSB / 6000 LSB/g
            = 0.0367 g

CRC
      1Ch
            CRC of 0500DCh, see section 5.2
```


6.1.1 加速数据转换示例

例如，若 ACC_X 寄存器读取结果为 ACC_X = 0500DC1Ch，则寄存器内容按如下方式转换为加速度：

OP[31: 26]		Data[23:8]				CRC[7:0]	
0	5	0	0	D	C	1	C

OP + RS

```
05h = 0000 0101b
      0000 01b          = 指令码 = 读取 ACC _X
      01b               = 返回状态（RS 位）= 无错误

数据 = ACC _X 寄存器内容
      00DCh
            00DCh → 220d      = 的二进制补码格式
            加速度：
            = 220 LSB / 灵敏度 (模式 1)
            = 220 LSB / 6000 LSB/g
            = 0.0367 克

CRC
      1Ch
            0500DCh 的 CRC，见第 5.2 节
```

6.1.2 Example of Temperature Data Conversion

For example, if TEMPERATURE register read results: TEMPERATURE = 15161E0Ah, the register content is converted to temperature as follows:

OP[31:26] + RS[25:24]		Data[23:8]				CRC[7:0]	
1	5	1	6	1	E	0	A

OP + RS

```
15h = 0001 0101b
      0001 01b            = OP code = Read TEMP
      01b                 = return status (RS bits) = no error

Data = TEMPERATURE register content
      161Eh
            161Eh → 5662d       = in 2's complement format
            Temperature:
            = -273 + (5662 / 18.9)
            = +26.6°C

CRC
      0Ah
            CRC of 15161Eh, see section 5.2
```


6.1.2 温度数据转换示例

例如，如果 TEMPERATURE 寄存器读取结果为 TEMPERATURE = 15161E0Ah，则寄存器内容按以下方式转换为温度：

OP[31: 26]		Data[23:8]				CRC[7:0]	
1	5	1	6	1	E	0	A

OP + RS

```
15h = 0001 0101b
      0001 01b            = OP code = Read TEMP
      01b                 = return status (RS bits) = no error

数据 = TEMPERATURE 寄存器内容
      161Eh
            161Eh → 5662d       = 的二进制补码格式
            温度：
            = -273 + (5662 / 18.9)
            = +26.6°C

CRC
      0Ah
            CRC of 15161Eh, see section 5.2
```

6.1.3 Example of Angle Data Conversion

Angle outputs must be enabled before angles can be read from registers. See section 6.6 for details.

For example, if ANG_X register read results: ANG_X = 250F8825h, the register content is converted to angle (degrees) as follows:

OP[31:26] + RS[25:24]		Data[23:8]				CRC[7:0]	
2	5	0	F	8	8	2	5

OP + RS

```
25h = 0010 0101b
      0010 01b            = OP code = Read ANG_X
      01b                 = return status (RS bits) = no error

Data = ANG_X register content
      0F88h
            0F88h → 3976d       = in 2's complement format
            Angle in degrees:
            = 3976/2^14*90
            = 21.84°

CRC
      25h
            CRC of 250F88h, see section 5.2
```


6.1.3 角度数据转换示例

在从寄存器读取角度之前，必须启用角度输出。有关详细信息，请参阅第 6.6 节。

例如，如果 ANG_X 寄存器读取结果为 ANG_X = 250F8825h，则寄存器内容按如下方式转换为角度（度）：

OP[31: 26]		Data[23:8]				CRC[7:0]	
2	5	0	F	8	8	2	5

OP + RS

```
25h = 0010 0101b
      0010 01b            = OP code = Read ANG_X
      01b                 = return status (RS bits) = no error

数据 = ANG_X 寄存器内容
      0F88h
            0F88h → 3976d       = 在补码格式下
            Angle in degrees:
            = 3976/2^14*90
            = 21.84°

CRC
      25h
            CRC of 250F88h, see section 5.2
```

6.2 STO

Table 21 STO (self-test output) description

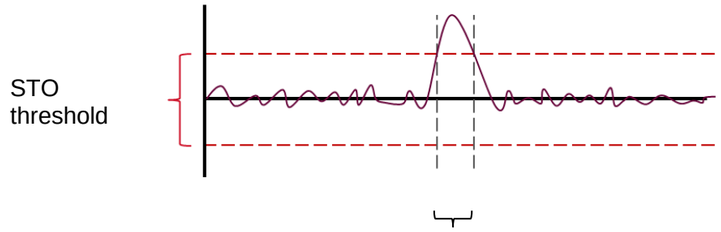
Bank	Addr	Name	No. of bits	Read / Write	Description
0 1	04h	STO	16	R	Self-test output in 2's complement format

Table 22 STO operation

Operation	SPI Frame	SPI Frame Hex
Read STO (self-test output)	0001 0000 0000 0000 0000 0000 1110 1001	100000E9h

If self-test option is desired in application, following guidelines should be taken into account. STO is used to monitor if accelerometer is functioning correctly. It provides information on signal saturation during vibration and shock events. STO should be read continuously in the normal operation sequence after XYZ acceleration readings.

STO threshold monitoring should be implemented on application software. Failure thresholds and failure tolerant time of the system are application specific and should be carefully validated. Monitoring can be implemented by counting the subsequent "STO signal exceeding threshold" –events. Examples for STO thresholds are shown in Table 23.



Component failure can be suspected if the STO signal exceeds the threshold level continuously after performing component hard reset in static (no vibration) condition.

Table 23 Examples for STO Thresholds

Mode	Full-scale	Examples for STO thresholds
1	± 1.2g	±1800 LSB
2	± 2.4g	±900 LSB
3	Inclination mode[3]	±3600 LSB
4	Inclination mode[3]	±3600 LSB

[3] Inclination mode. Dynamic range is dependent on orientation in gravity. See 2.11.1

6.2 STO

表 21 STO（自测试输出）说明

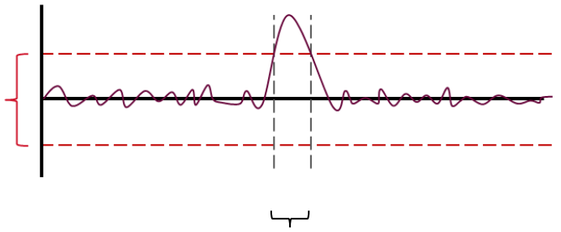
银行地			位数	阅读 / 编写	描述
0 1	04 小时		16	R	2 的补码格式下的自检输出

表 22 STO 操作

操作	SPI 帧	SPI 帧十六进制
读取 STO（自检输出）	0001 0000 0000 0000 0000 0000 1110 1001	100000E9h

如果应用程序需要自测功能，请遵循以下指南。STO 用于监控加速度计是否正常工作。它在振动和冲击事件期间提供信号饱和信息。STO 应在 XYZ 加速度读数之后，在正常操作序列中连续读取。

STO 阈值监控应在应用软件中实施。系统的故障阈值和故障容忍时间因应用而异，应仔细验证。监控可通过统计后续的"STO 信号超过阈值"事件来实现。STO 阈值的示例见表 23。



如果在对组件执行硬复位后，在静态（无振动）条件下 STO 信号持续超过阈值水平，则可能怀疑组件故障。

表 23 STO 阈值的示例

Mode	全尺寸	STO 阈值的示例
1	± 1.2g	±1800 最低有效位
2	± 2.4g	±900 最低有效位
3	倾角模式[3]	±3600 最低有效位
4	倾角模式[3]	±3600 最低有效

[3] 倾角模式。动态范围取决于重力方向。参见 2.11.1

6.2.1 Example of Self-Test Analysis

For example, if STO register read results: STO = 1100017Bh, the register value can be converted as follows:

OP[31:26] + RS[25:24]		Data[23:8]				CRC[7:0]	
1	1	0	0	0	1	7	B

OP + RS

```
      11h = 0001 0001b
            0001 00b            = OP code = Read STO
            01b                 = return status (RS bits) = no error

Data = STO register content
      0001h
            0001h → 1d          = in 2's complement format
            Self-test reading:
            = 1
            See Table 12 for recommended STO threshold values
CRC
      7Bh
      CRC of 110001h, see section 5.2
```


6.2.1 自检分析示例

例如，如果 STO 寄存器读取结果为：STO = 1100017Bh，则寄存器值可按如下方式转换：

OP[31:26] + RS[25:24]		Data[23:8]				CRC[7:0]	
1	1	0	0	0	1	7	B

OP + RS

```
      11h = 0001 0001b
            0001 00b            = OP code = Read STO
            01b                 = return status (RS bits) = no error

数据 = STO 寄存器内容
      0001h
            0001h → 1d          = 的二进制补码格式
            自测阅读：
            = 1
            参见表 12 获取推荐的 STO 阈值
CRC
      7Bh
      CRC of 110001h, see section 5.2
```

6.3 STATUS

Table 24 STATUS description

Bank	Addr	Name	No. of bits	Read / Write	Description
0 1	06h	STATUS	16	R	Status Summary combining ERR_FLAG1 and ERR_FLAG2

Table 25 STATUS operation

Operation	SPI Frame	SPI Frame Hex
Read Status Summary	0001 1000 0000 0000 0000 0000 1110 0101	180000E5h

Table 26 STATUS register

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Bit
Reserved						DIGI1	DIGI2	CLK	SAT	TEMP_SAT	PWR	MEM	PD	MODE_CHANGE	PIN_CONTINUITY	Read

6.3 状态

表 24 STATUS 描述

银行地			位数	阅读 / 编写	描述
0 1	06h		16	R	结合 ERR_FLAG1 和 ERR FLAG2_ 的状态摘要

表 25 STATUS 操作

操作	SPI 帧	SPI 帧十六进制
读取状态摘要	0001 1000 0000 0000 0000 0000 1110 0101	180000E5h

表 26 状态寄存器

D15	D14					D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Bit
保留																Read

Table 27 STATUS register bit description

Bit	Name	Description	Required action/explanation
9	DIGI1	Digital block error type 1	SW or HW reset needed
8	DIGI2	Digital block error type 2	SW or HW reset needed
7	CLK	Clock error	SW or HW reset needed
6	SAT	Signal saturated in signal path	Acceleration too high and acceleration and inclination readings not usable. Component failure possible. All acceleration, inclination, and STO output data is invalid.
5	TEMP_SAT	Temperature signal path saturated	External temperature too high or low. Component failure possible
4	PWR	Start-up indication or Voltage level failure	[After star-up or reset] This flag is set high. No actions needed.
			[During normal operation] External voltages too high or low. Component failure possible. SW or HW reset needed.
3	MEM	Error in non-volatile memory	Memory check failed. Possible component failure SW or HW reset needed.
2	PD	Device in power down mode	If power down is not requested. SW or HW reset needed
1	MODE_CHANGE	Operation mode changed	Bit is set high if operation mode has been changed If mode change is not requested SW or HW reset needed
0	PIN_CONTINUITY	Component internal connection error	Possible component failure

Software (SW) reset is done with SPI operation (see 5.1.4). Hardware (HW) reset is done by power cycling the sensor. If these do not reset the error, then possible component error has occurred and system needs to be shut down and part returned to supplier.

表 27 STATUS 寄存器位描述

Bit	Name	描述	必填操作/说明
9	DIGI1	数字块错误类型 1	需要软件或硬件复位
8	DIGI2	数字块错误类型 2	需要软件或硬件复位
7	CLK	时钟误差	需要软件或硬件复位
6	SAT	信号路径中信号饱和	加速度过高，加速度和倾斜读数不
5	TEMP SAT_	温度信号路径饱和	外部温度过高或过低。可能发生组件
4	PWR	启动指示或电压电平 失败	[启动或复位后] 此标志置高。无
			[正常运行期间] 外部电压过高或过 需要软件或硬件复位。
3	MEM	非易失性存储器中的错误	内存检查失败。可能存在组件故障。 需要软件或硬件复位。
2	PD	设备处于关机模式	如果未请求关机。 需要软件或硬件复位
1	模式切换_	操作模式已更改	如果操作模式已更改，则 Bit 设置 如果未请求模式切换 需要软件或硬件复位
0	PIN_连续性	组件内部连接错误 可能的组件故障	

软件（SW）复位通过 SPI 操作完成（参见 5.1.4）。硬件（HW）复位通过对传感器进行电源循环完成。如果这些操作无法复位错误，则可能发生了组件故障，此时需要关闭系统并将部件退回供应商。

6.3.1 Example of STATUS summary reset

STATUS summary is reset by reading it. Below is an example of MOSI commands and corresponding MISO responses for command Read STATUS summary when there is SAT bit high in STATUS summary (Data = 0x0040).

Due to off-frame protocol of SPI the first response to MOSI command is a response to earlier MOSI command and is thus not applicable in this example.

The Return Status bits show an error (b'11) even with the first MOSI command and are reset after the second command (b'01). Return Status bits are defined in Chapter 5.1.5.

#	MOSI command	MISO response	Return Status bits (RS)	Data
1	0x180000E5	*don't care*	b'11	*don't care*
2	0x180000E5	0x1b00407a	b'11	0x0040
3	0x180000E5	0x19004079	b'01	0x0040
4	0x180000E5	0x1900006a	b'01	0x0000

6.3.1 STATUS 摘要重置示例

STATUS 摘要通过读取其值进行重置。以下是 SAT 位为高时执行"Read STATUS summary"命令的 MOSI 命令及对应的 MISO 响应示例（数据 = 0x0040）。

由于 SPI 的离帧协议，对 MOSI 命令的第一个响应是之前 MOSI 命令的响应，因此在此示例中不适用。

返回状态位显示错误（b'11），即使在第一个 MOSI 命令下也是如此，并在第二个命令后重置（b'01）。返回状态位在 5.1.5 章中定义。

#	MOSI 命令	MISO 响应	返回状态数据位 (	
1	0x180000E5	不在乎	b'11	不在乎
2	0x180000E5	0x1b00407a	b'11	0x0040
3	0x180000E5	0x19004079	b'01	0x0040
4	0x180000E5	0x1900006a	b'01	0x0000

6.4 Error Flag Block

Table 28 Error flag block description

Bank	Addr	Register Name	No. of bits	Read / Write	Description
0	07h	ERR_FLAG1	16	R	Error flags
0	08h	ERR_FLAG2	16	R	Error flags

Table 29 Error flag block operations

Operation	SPI Frame	SPI Frame Hex
Read ERR_FLAG1	0001 1100 0000 0000 0000 0000 1110 0011	1C0000E3
Read ERR_FLAG2	0010 0000 0000 0000 0000 0000 1100 0001	200000C1h

STATUS register contains combination of the information in the ERR_FLAG1 and ERR_FLAG2 registers; if there is an error, it is reflected in STATUS. ERR_FLAG registers can be used to further assess reason for error. Note that reading ERR_FLAG registers does not reset error flags in STATUS register nor reset RS bits.

6.4.1 ERR_FLAG1

Table 30 ERR_FLAG1 register

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Bit
Reserved				ADC_SAT	AFE_SAT										MEM	Read

Table 31 ERR_FLAG1 register bit description

Bit	Name	Description
15:12	Reserved	Reserved
11	ADC_SAT	Signal saturated at A2D
10:1	AFE_SAT	Signal saturated at C2V
0	MEM	Error in non-volatile memory

6.4 错误标志块

表 28 错误标志块说明

银行地		注册名称	位数	读取 /	描述
0	07h	ERR FLAG1_	16	R	错误标志
0	08h	ERR FLAG2_	16	R	错误标志

表 29 错误标志块操作

操作	SPI 帧	SPI 帧十六进制
读取 ERR FLAG1_	0001 1100 0000 0000 0000 0000 1110 0011	1C0000E3
查看 ERR FLAG2_	0010 0000 0000 0000 0000 0000 1100 0001	200000C1h

STATUS 寄存器包含 ERR_FLAG1 和 ERR_FLAG2 寄存器中的信息组合；如果发生错误，则会在 STATUS 中反映。ERR¯FLAG 寄存器可用于进一步评估错误原因。注意，读取 ERR¯FLAG 寄存器不会重置 STATUS 寄存器中的错误标志位，也不会重置 RS 位。

6.4.1 ERR FLAG1_

表 30 ERR_FLAG1 寄存器

D15	D14					D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Bit
保留					AFE_SAT											Read

表 31 ERR_FLAG1 寄存器位描述

Bit	Name	描述
15:12	保留	保留
11	ADC SAT_	信号在 A2D 处饱和
10:1	AFE SAT_	C2V 处信号饱和
0	MEM	非易失性存储器中的错误

6.4.2 ERR_FLAG2

Table 32 ERR_FLAG2 register

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Bit
Reserved	D_EXTC	A_AEXT_C	AGND	VDD	Reserved	MODE_CHANGE	PD	MEMORY_CRC	Reserved	APWR	DPWR	REFV	APWR_2	TEMP_SAT	CLK	Read

Table 33 ERR_FLAG2 register bit description

Bit	Name	Description
15	Reserved	Reserved
14	D_EXT_C	External capacitor connection error
13	A_EXT_C	External capacitor connection error
12	AGND	Analog ground connection error
11	VDD	Supply voltage error
10	Reserved	Reserved
9	MODE_CHANGE	Operation mode changed by user
8	PD	Device in power down mode
7	MEMORY_CRC	Memory CRC check failed
6	Reserved	Reserved
5	APWR	Analog power error
4	DPWR	[After star-up or reset] This flag is set high. No actions needed. [During normal operation] Digital power error. Component failure possible. SW or HW reset needed.
3	VREF	Reference voltage error
2	APWR_2	Analog power error
1	TEMP_SAT	Temperature signal path saturated
0	CLK	Clock error

6.4.2 ERR FLAG2_

表 32 ERR_FLAG2 寄存器

D15	D14					D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Bit
																Read

表 33 ERR_FLAG2 寄存器位描述

Bit	Name	描述
15	保留	保留
14	D_EXT_C	外部电容连接错误
13	A_EXT_C	外部电容连接错误
12	AGND	模拟地连接错误
11	VDD	供电电压误差
10	保留	保留
9	模式切换_	用户更改的操作模式
8	PD	设备处于关机模式
7	内存 CRC_	内存 CRC 校验失败
6	保留	保留
5	APWR	模拟电源错误
4	DPWR	[启动或重置后] 此标志已置高。无需操作。 [正常运行期间] 数字电源错误。可能发生组件故障。 需要软件或硬件复位。
3	VREF	参考电压误差
2	APWR 2_	模拟电源错误
1	TEMP SAT_	温度信号路径饱和
0	CLK	时钟误差

6.5 CMD

Table 34 CMD description

Bank	Addr	Register Name	No. of bits	Read / Write	Description
0	0Dh	CMD	16	R / W	Sets operation mode, SW Reset and Power down mode

Table 35 CMD operations

Command	SPI Frame	SPI Frame hex
Read CMD	0011 0100 0000 0000 0000 0000 1101 1111	340000DFh
Change to mode1	1011 0100 0000 0000 0000 0000 0001 1111	B400001Fh
Change to mode2	1011 0100 0000 0000 0000 0001 0000 0010	B4000102h
Change to mode3	1011 0100 0000 0000 0000 0010 0010 0101	B4000225h
Change to mode4	1011 0100 0000 0000 0000 0011 0011 1000	B4000338h
Set power down mode	1011 0100 0000 0000 0000 0100 0110 1011	B400046Bh
Wake up from power down mode	1011 0100 0000 0000 0000 0000 0001 1111	B400001Fh
SW Reset	1011 0100 0000 0000 0010 0000 1001 1000	B4002098h

Table 36 CMD register

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Bit
Reserved								Factory use	Factory use	SW_RST	Factory use	Factory use	PD	MODE		Read

Table 37 CMD register bit description

Bit	Name	Description
15:8	Reserved	Reserved
7	Factory use	Factory use
6	Factory use	Factory use
5	SW_RST	Software (SW) Reset
4	Factory use	Factory use
3	Factory use	Factory use
2	PD	Power Down
1:0	MODE	Operation Mode

Sets operation mode of the SCL3300-D01. After power-off, reset (SW or HW), power down mode or unintentional power-off, normal start-up sequence must be followed. Note: mode will be set to default mode1.

Operation modes are described in section 4.3.

Changing mode will set Status Summary bit 1 to high, setting / waking up from power down mode will set Status Summary bit 2 to high (see 6.3.) Thus RS bits will show '11' (see 5.1.5.). **Note**: User must not configure other than given valid commands, otherwise power-off, reset, or power down is required.

6.5 CMD

表 34 CMD 描述

银行地		注册名称	编号 bits	阅读 / 编写	描述
0	0Dh	CMD	16	R / W	

表 35 CMD 操作

命令	SPI 帧	SPI 帧十六进制
读取 CMD	0011 0100 0000 0000 0000 0000 1101 1111	340000DFh
切换到模式 1	1011 0100 0000 0000 0000 0000 0001 1111	B400001Fh
切换到模式 2	1011 0100 0000 0000 0000 0001 0000 0010	B4000102h
切换到模式 3	1011 0100 0000 0000 0000 0010 0010 0101	B4000225h
切换到模式 4	1011 0100 0000 0000 0000 0011 0011 1000	B4000338h
设置关机模式	1011 0100 0000 0000 0000 0100 0110 1011	B400046Bh
从关机模式唤醒		B400001Fh
SW 复位	1011 0100 0000 0000 0010 0000 1001 1000	B4002098h

表 36 CMD 寄存器

D15	D14					D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Bit
																Read

表 37 CMD 寄存器位描述

Bit	Name	描述
15:8	保留	保留
7	工厂使用	工厂使用
6	工厂使用	工厂使用
5	SW RST_	软件（SW）复位
4	工厂使用	工厂使用
3	工厂使用	工厂使用
2	PD	电源关闭
1:0	MODE	运行模式

设置 SCL3300-D01 的工作模式。断电后，复位（软件或硬件）、关机模式或非预期断电后，必须遵循正常启动序列。注意：模式将设置为默认模式 1。

操作模式在第 4.3 节中描述。

切换模式会将状态摘要位 1 置为高电平，设置/从掉电模式唤醒会将状态摘要位 2 置为高电平（参见 6.3）。因此 RS 位将显示"11"（参见 5.1.5）。注意：用户不得配置除给定有效命令外的其他内容，否则需要执行掉电、复位或掉电操作。

6.6 ANG_CTRL

Table 38 ANG_CTRL description

Bank	Addr	Register Name	No. of bits	Read / Write	Description
0	0Ch	ANG_CTRL	16	W	Enable angle outputs.

Table 39 ANG_CTRL operations

Command	SPI Frame	SPI Frame hex
Enable Angle Outputs	1011 0000 0000 0000 0001 1111 0110 1111	B0001F6Fh

Table 40 ANG_CTRL register

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Bit
Not Used [15:5]											Enable Angle Outputs [4:0] = b'11111					Write

ANG_CTRL is a 5-bit registers to control angle outputs. Default value for register is 00h. Angle outputs are enabled by writing 1Fh to ANG_CTRL.

6.7 WHOAMI

Table 41 WHOAMI description

Bank	Addr	Register Name	No. of bits	Read / Write	Description
0	10h	WHOAMI	8	R	8-bit register for component identification

Table 42 WHOAMI operations

Operation	SPI Frame	SPI Frame Hex
Read WHOAMI	0100 0000 0000 0000 0000 0000 1001 0001	40000091h

Table 43 WHOAMI register

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Bit
								-	-	-	-	-	-	-	-	Write
Not Used [15:8]								Component ID [7:0] = C1h								Read

WHOAMI is a 8-bit register for component identification. Returned value is C1h.

Note: as returned value is fixed, this can be used to ensure SPI communication is working correctly.

6.6 ANG CTRL_

表 38 ANG_CTRL 描述

银行地		注册名称	编号 bits	阅读 / 编写	描述
0	0Ch	ANG CTRL_	16	W	启用角度输出。

表 39 ANG_CTRL 操作

命令	SPI 帧	SPI 帧十六进制
启用角度输出	1011 0000 0000 0000 0001 1111 0110 1111	B0001F6Fh

表 40 ANG_CTRL 寄存器

D15	D14					D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Bit
未使用 [15:5]											启用角度输出 [4:0] = b'11111					编写

ANG_CTRL 是一个 5 位寄存器，用于控制角度输出。寄存器的默认值为 00h。通过将 1Fh 写入 ANG_CTRL 来启用角度输出。

6.7 我是谁

表 41 WHOAMI 描述

银行地		注册名称	编号 bits	阅读 / 编写	描述
0	10h	我是谁	8	R	8 位寄存器用于组件识别

表 42 WHOAMI 操作

操作	SPI 帧	SPI 帧十六进制
阅读 WHOAMI	0100 0000 0000 0000 0000 0000 1001 0001	40000091h

表 43 WHOAMI 寄存器

D15	D14					D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	
								-	-	-	-	-	-	-	-	编写
未使用 [15:8]								组件 ID [7:0] = C1h								Read

WHOAMI 是用于组件识别的 8 位寄存器。返回值为 C1h。

注意：由于返回值是固定的，这可用于确保 SPI 通信工作正常。

6.8 Serial Block

Table 44 Serial block description

Bank	Addr	Register Name	No. of bits	Read / Write	Description
1	19h	SERIAL1	16	R	Component serial part 1
1	1Ah	SERIAL2	16	R	Component serial part 2

Table 45 Serial block operations

Operation	SPI Frame	SPI Frame Hex
Read SERIAL1	0110 0100 0000 0000 0000 0000 1010 0111	640000A7h
Read SERIAL2	0110 1000 0000 0000 0000 0000 1010 1101	680000ADh

Serial Block contains sensor serial number in two 16 bit registers in register bank #1, see 6.5 CMD for information how to switch register banks. The same serial number is also written on top of the sensor.

The following procedure is recommended when reading serial number:

1. Change active register bank to #1
2. Read registers 19h and 1Ah
3. Change active register back to bank #0
4. Resolve serial number:
    1. Combine result data from 1Ah[16:31] and 19h[0:15]
    2. Convert HEX to DEC
    3. Add letters “B33” to end

6.8 序列块

表 44 串行块描述

银行地		注册名称	位数	读取 /	描述
1	19h	SERIAL1	16	R	组件序列部分 1
1	1Ah	SERIAL2	16	R	组件序列部分 2

表 45 串行块操作

操作	SPI 帧	SPI 帧十六进制
读取 SERIAL1	0110 0100 0000 0000 0000 0000 1010 0111	640000A7h
读取 SERIAL2	0110 1000 0000 0000 0000 0000 1010 1101	680000ADh

序列块在寄存器组#1 的两个 16 位寄存器中包含传感器序列号，有关如何切换寄存器组的详细信息，请参阅 6.5 CMD。相同的序列号也刻在传感器顶部。

读取序列号时，建议按以下步骤操作：

1. 将活动寄存器组更改为 #1
2. 读取寄存器 19h 和 1Ah
3. 将活动寄存器改回 0 号寄存器
4. 解析序列号：
    1. 合并 1Ah[16:31] 和 19h[0:15] 的结果数据
    2. 将十六进制转换为十进制
    3. 在末尾添加字母"B33"

6.8.1 Example of Resolving Serial Number

1 Change active register bank to #1
SPI Request SWITCH_TO_BANK_1
Request: FC00016E
Response: XXXXXXXX, response to previous command

2. Read registers 19h and 1Ah
SPI Request READ_SERIAL1:
Request: 640000A7
Response: FD0001E1, response to switch command

SPI Request READ_SERIAL2:
Request: 680000AD
Response: 65F7DA19, response to serial1, data: F7DA

3. Change active register back to bank #0
SPI Request SWITCH_TO_BANK_0
Request: FC000073
Response: 693CE54F, response to serial2, data: 3CE5

4. Resolve serial number
1. Combined Serial number: 3CE5F7DA
2. HEX to DEC: 1021704154
3. Add “B33”: 1021704154B33

➔ Full Serial number: 1021704154B33

6.8.1 序列号解析示例

1 将活动寄存器组更改为 #1
SPI 请求 SWITCH_TO_BANK_1
请求：FC00016E
响应：XXXXXXXX，对先前命令的响应

2. 读取寄存器 19h 和 1Ah
SPI 请求 READ_SERIAL1:
请求：640000A7
响应：FD0001E1，对切换命令的响应

SPI 请求 READ_SERIAL2:
请求：680000AD
响应：65F7DA19，对 serial1 的响应，数据：F7DA

3. 将活动寄存器改回 0 号银行
SPI 请求 SWITCH_TO_BANK_0
请求：FC000073
响应：693CE54F，对 serial2 的响应，数据：3CE5

4. 解析序列号
1. 组合序列号：3CE5F7DA
2. 十六进制转十进制：1021704154
3. 添加“B33”：1021704154B33

➔ 完整序列号：1021704154B33

6.9 SELBANK

Table 46 SELBANK description

Bank	Addr	Register Name	No. of bits	Read / Write	Description
0 1	1Fh	SELBANK	16	R	Switch between active register banks

Table 47 SELBANK operations

Command	SPI Frame	SPI Frame hex
Read current bank	0111 1100 0000 0000 0000 0000 1011 0011	7C0000B3h
Switch to bank #0	1111 1100 0000 0000 0000 0000 0111 0011	FC000073h
Switch to bank #1	1111 1100 0000 0000 0000 0001 0110 1110	FC00016Eh

SELBANK is used to switch between memory banks #0 and #1. It's recommended to keep memory bank #0 selected unless register from bank #1 is required, for example, reading serial number of sensor. After using bank #1 user should switch back to bank #0.

7 Application information

7.1 Application Circuitry and External Component Characteristics

See Figure 16 and Table 48 for specification of the external components. The PCB layout example is shown in Figure 17.

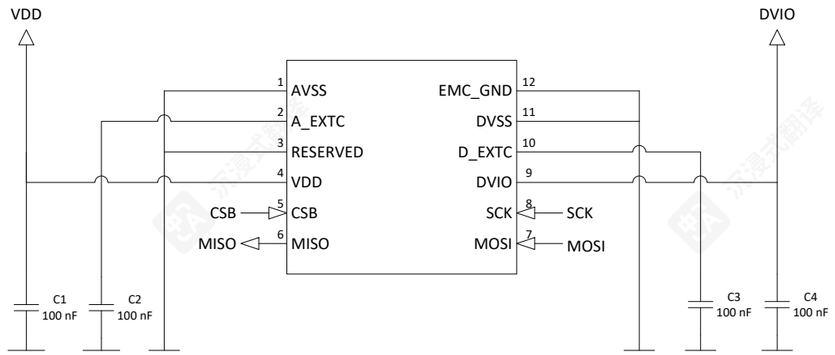


Figure 16 Application schematic.

6.9 SELBANK

表 46 SELBANK 描述

Bank	Addr	注册名称	编号 bits	阅读 / 编写	描述
0 1	1Fh	SELBANK	16	R	在活跃寄存器组之间切换

表 47 SELBANK 操作

命令	SPI 帧	SPI 帧十六进制
读取当前银行	0111 1100 0000 0000 0000 0000 1011 0011	7C0000B3h
切换到银行 #0	1111 1100 0000 0000 0000 0000 0111 0011	FC000073h
切换到银行 #1	1111 1100 0000 0000 0000 0001 0110 1110	FC00016Eh

SELBANK 用于在内存银行#0 和#1 之间切换。除非需要从银行#1 读取寄存器（例如读取传感器的序列号），否则建议保持内存银行#0 被选中。使用完银行#1 后，应切换回银行#0。

7 应用信息

7.1 应用电路与外部元件特性

参见图 16 和表 48 以了解外部组件的规格。PCB 布局示例如图 17 所示。

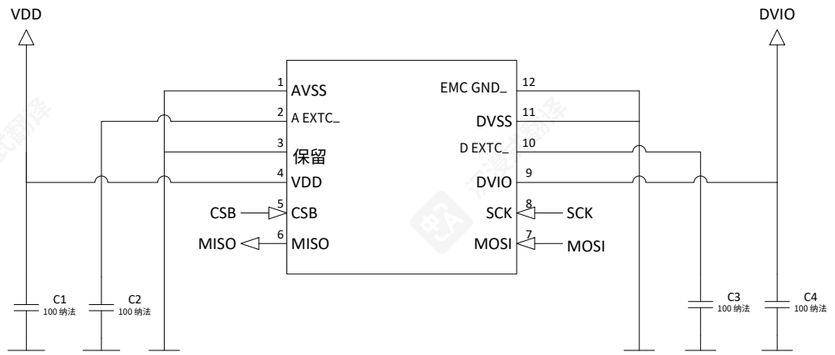


图 16 应用示意图。

Table 48 External component description for SCL3300-D01.

Symbol	Description	Min.	Nom.	Max.	Unit
C1	Decoupling capacitor between VDD and GND Recommended component: Murata GCM155R71C104KA55, 0402, 16V, X7R ESR Please confirm capacitor availability from www.murata.com	70	100	130 100	nF mΩ
C2	Decoupling capacitor between A_EXTC and GND Recommended component: Murata GCM155R71C104KA55, 0402, 16V, X7R ESR Please confirm capacitor availability from www.murata.com	70	100	130 100	nF mΩ
C3	Decoupling capacitor between D_EXTC and GND Recommended component: Murata GCM155R71C104KA55, 0402, 16V, X7R ESR Please confirm capacitor availability from www.murata.com	70	100	130 100	nF mΩ
C4	Decoupling capacitor between DVIO and GND Recommended component: Murata GCM155R71C104KA55, 0402, 16V, X7R ESR Please confirm capacitor availability from www.murata.com	70	100	130 100	nF mΩ

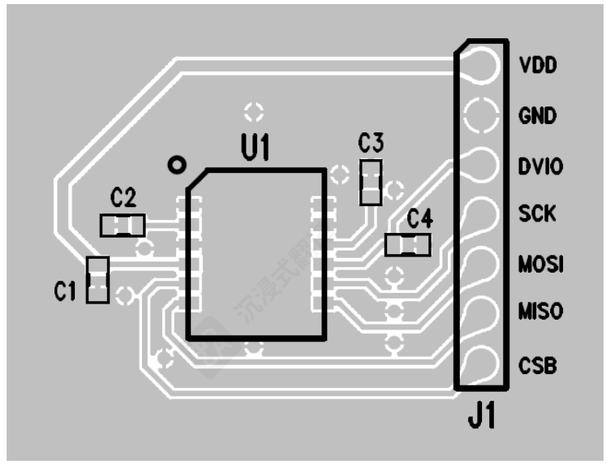


Figure 17. Application PCB layout.

General circuit diagram and PCB layout recommendations for SCL3300-D01:

1. Connect decoupling SMD capacitors (C1 - C4) right next to respective component pins.
2. Place ground plate under component.
3. Do not route signals or power supplies under the component on top layer.
4. Ensure good ground connection of DVSS, AVSS and EMC_GND pins

表 48 SCL3300-D01 外部组件说明。

符号 说明		Min.	Nom.	Max.	Unit
C1	VDD 与 GND 之间的去耦电容 推荐元件：Murata GCM155R71C104KA55，0402 ESR	70	100	130 100	nF mΩ
C2	A_EXTC 与 GND 之间的解耦电容 推荐元件：Murata GCM155R71C104KA55，0402 ESR	70	100	130 100	nF mΩ
C3	D_EXTC 与 GND 之间的解耦电容 推荐元件：Murata GCM155R71C104KA55，0402 ESR	70	100	130 100	nF mΩ
C4	DVIO 与 GND 之间的解耦电容 推荐元件：Murata GCM155R71C104KA55，0402 ESR	70	100	130 100	nF mΩ

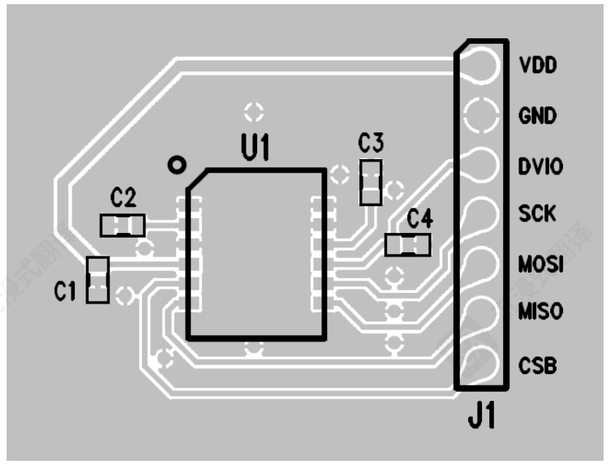


图 17。应用 PCB 布局。

SCL3300-D01 的通用电路图及 PCB 布局建议：

1. 在相应元件引脚旁直接连接去耦 SMD 电容（C1 - C4）。
2. 将接地板置于元件下方。
3. 顶层不得在元件下方布设信号或电源。
4. 确保 DVSS、AVSS 和 EMC_GND 引脚的良好接地连接

7.2 Assembly Instructions

The Moisture Sensitivity Level of the component is Level 3 according to the IPC/JEDEC JSTD-020C. The part is delivered in a dry pack. The manufacturing floor time (out of bag) at the customer's end is 168 hours.

Usage of PCB coating materials may penetrate component lid and affect component performance. PCB coating is not allowed.

Sensor components shall not be exposed to chemicals which are known to react with silicones, such as solvents. Sensor components shall not be exposed to chemicals with high impurity levels, such as Cl-, Na+, NO3-, SO4-, NH4+ in excess of >10 ppm. Flame retardants such as Br or P containing materials shall be avoided in close vicinity of sensor component. Materials with high amount of volatile content should also be avoided.

If heat stabilized polymers are used in application, user should check that no iodine, or other halogen, containing additives are used.

For additional assembly related details please refer to technical note Assembly instructions of Dual Flat Lead Package (DFL).

APP 2702 Rev.2 Assembly_Instructions_for_DFL_Package

8 Frequently Asked Questions

- How can I be sure SPI communication is working?
  - Read register WHOAMI (10h), the response should be C1h.
- Why do I get wrong results when I read data?
  - SCL3300-D01 uses off-frame protocol (see 5.1.2 Protocol), make sure to utilize this correctly.
  - Confirm time between SPI requests (CSB high) is at least 10 µs.
  - Ensure SCL3300-D01 is correctly started (see 4.2 Start-up sequence).
  - Read RS bits (see 5.1.5 Return Status), if error is shown read Status Summary (see 6.3 STATUS) for further information.
  - Confirm correct sensitivity is used for current operation mode (see 4.3 Operation modes)
- Why all angle outputs read only zeroes?
  - Ensure that angle outputs are enabled after startup (see 4.2 Start-up sequence and 6.6 ANG_CTRL)

7.2 组装说明

该元件的湿气敏感等级为 IPC/JEDEC JSTD-020C 标准中的 3 级。该元件以干燥包装形式交付。客户端的制造地板时间（拆包后）为 168 小时。

PCB 涂覆材料的使用可能渗透至元器件顶部并影响其性能。PCB 涂覆不允许。

传感器部件不应接触已知会与硅橡胶发生反应的化学品，例如溶剂。传感器部件不应接触高杂质含量的化学品，例如 Cl-、Na+、NO3-、SO4-、NH4+，其含量超过 >10 ppm。应避免在传感器部件附近使用含 Br 或 P 的阻燃材料。还应避免使用高挥发含量的材料。

如果应用中使用热稳定聚合物，用户应检查是否未使用含碘或其他卤素的添加剂。

有关更多封装相关细节，请参阅技术说明《双扁平引线封装（DFL）的组装说明》。

APP 2702 Rev.2 组装_说明_用于_DFL_套件

8 常见问题

- 如何确保 SPI 通信正常工作?
  - 读取寄存器 WHOAMI (10h)，响应应为 C1h。
- 为什么读取数据时会得到错误结果?
  - SCL3300-D01 使用非帧协议（参见 5.1.2 协议），请确保正确使用。
  - 确认 SPI 请求（CSB 高）之间的时间间隔至少为 10 µs。
  - 确保 SCL3300-D01 已正确启动（参见 4.2 启动流程）。
  - 读取 RS 位（参见 5.1.5 返回状态），如果显示错误，请读取状态摘要（参见 6.3 状态）以获取更多信息。
  - 确认当前操作模式（见 4.3 操作模式）使用了正确的灵敏度。
- 为什么所有角度输出都只读零?
  - 确保在启动后启用角度输出（参见 4.2 启动序列和 6.6 ANG_CTRL）

9 Order Information

Order Code	Description	Packing	Qty
SCL3300-D01-004	3-axis inclinometer with digital SPI interface	Bulk	4pcs
SCL3300-D01-1	3-axis inclinometer with digital SPI interface	T&R	100pcs
SCL3300-D01-10	3-axis inclinometer with digital SPI interface	T&R	1000pcs

47 (47)

9 订单信息

订单编号	描述	包装	Qty
SCL3300-D01-004	三轴数字 SPI 接口倾角仪	Bulk	4pcs
SCL3300-D01-1	三轴倾角仪，带数字 SPI 接口	T&R	100 件
SCL3300-D01-10	三轴倾角仪，带数字 SPI 接口	T&R	1000 件